Autoregressive Seq2Seq Transformers for Risk-Managing Deep Reinforcement Learning in Delta-Neutral Option Trading via Cross-Attention Mechanisms

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Abstract

This project addresses the high-dimensional combinatorial challenge of delta-neutral options trading, where an agent must simultaneously evaluate multiple expiries, strikes, and instrument types (Calls/Puts) to execute optimal Buy, Sell, or Hold actions. We propose an Autoregressive Seq2Seq Transformer architecture that leverages cross-attention mechanisms to navigate this vast decision space. The framework integrates multi-modal data—including historical stock technicals, a deterministic 80-option chain, and real-time portfolio Greeks—into a unified latent representation. By utilizing an encoder-decoder structure, the model generates trade orders as sequential triplets, effectively mapping complex market states to multi-leg strategies. The agent is trained using Proximal Policy Optimization (PPO) with a specialized delta-neutrality reward penalty ($\lambda = 100.0$).

Central to this research is the hypothesis that the reinforcement learning agent will autonomously develop a structural financial understanding of the option chain without explicit encoding. We evaluate this through mechanistic interpretability of cross-attention heatmaps, investigating three emergent signatures: Instrument Type Awareness (bilateral symmetry between Calls and Puts), Expiry Discrimination (dynamic attention shifting between near and far expiry blocks), and Moneyness Approximation (autonomous tracking of the At-The-Money region). Experimental results demonstrate the framework's capacity to function as an implicit option chain parser, learning the structural language of derivatives markets while maintaining a balanced risk profile in a custom-built trading simulation.

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List of Acronyms

General AI and Machine Learning

- AI: Artificial Intelligence
- ML: Machine Learning
- LLM: Large Language Model
- NLP: Natural Language Processing
- RL: Reinforcement Learning
- PPO: Proximal Policy Optimization
- Seq2Seq: Sequence-to-Sequence
- GAE: Generalized Advantage Estimation
- GPT: Generative Pre-Trained Transformer
- Project Specific Architecture
- ATR: Autoregressive Trade Model
- [CLS_PORT]: Portfolio Summary Token (Classification-style token for portfolio state)
- SOS: Start of Sequence
- EOS: End of Sequence
- QKV: Query, Key, Value (Attention mechanism components)

Finance and Options Trading

- NIFTY: National Index Fifty (NIFTY 50 Index)
- LTP: Last Traded Price
- PnL: Profit and Loss
- GBM: Geometric Brownian Motion
- ATM: At-The-Money
- ITM: In-The-Money
- OTM: Out-Of-The-Money
- MDD: Maximum Drawdown
- RSI: Relative Strength Index
- MFI: Money Flow Index
- ROC: Rate of Change
- DTE: Days to Expiry

Hardware and Software Environment

- GPU: Graphics Processing Unit
- CUDA: Compute Unified Device Architecture
- MPS: Metal Performance Shaders
- VRAM: Video Random Access Memory (GPU memory)
- API: Application Programming Interface

Chapter 1

1.1 Introduction

The landscape of financial derivatives trading has been transformed by the advent of algorithmic systems capable of processing vast amounts of data in real-time. Among these derivatives, options represent one of the most complex instruments due to their non-linear payoff structures and sensitivity to multiple market variables. For institutional investors and market makers, the primary objective is often not just profit maximization, but the maintenance of a delta-neutral portfolio—a state where the overall value is immunized against small fluctuations in the underlying asset's price.

The Combinatorial Challenge

Options trading presents a massive combinatorial problem. For any given underlying asset, a trader must choose from a variety of variables:

- Expiries: Ranging from near-term to far-term monthly cycles.
- Strikes: Multiple price levels centered around the At-The-Money (ATM) price.
- Instrument Types: Call options (CE) and Put options (PE).
- Actions: Buying (long), selling (short), or holding existing positions.

In this project, the agent manages a deterministic chain of 80 options per timestep. Selecting the optimal combination of these variables to achieve a specific risk profile creates a search space that is computationally taxing for traditional rule-based systems or standard Reinforcement Learning (RL) architectures.

The Role of Sequence-to-Sequence Transformers

Traditional RL agents often struggle with high-dimensional action spaces. To overcome this, this research implements an Autoregressive Seq2Seq Transformer. By treating a trade order as a sequence of tokens—specifically triplets of Action, Strike, and Lot Size—the model generates complex, multi-leg strategies while conditioning each decision on the previous ones.

The architecture utilizes cross-attention mechanisms to bridge the gap between market state and action. The encoder processes a multi-modal sequence of historical stock technicals, theoretical option Greeks ($\Delta, \Gamma, \Theta, Vega, Rho$), and a "Rich Portfolio Summary". This research posits that the cross-attention mechanism functions as an implicit option chain parser, learning the structural language of derivatives markets—specifically instrument symmetry, expiry discrimination, and moneyness—through interaction alone.

Risk-Managed Reinforcement Learning and Mechanistic Interpretability

Unlike "black-box" profit seekers, the agent developed in this study is trained to be risk-aware. Using Proximal Policy Optimization (PPO), the agent is incentivized through a shaped reward function that includes a heavy penalty ($\lambda = 100.0$) for deviating from a delta-neutral state beyond a 0.5 threshold.

Beyond financial performance, this study emphasizes mechanistic interpretability. By analyzing the internal attention weights of the transformer decoder, the research evaluates whether the agent autonomously develops a financial understanding of the option chain. This project, submitted as part of the Master's in Computer Engineering curriculum at the University of Bridgeport, aims to demonstrate that modern NLP-inspired architectures can provide a robust framework for both solving and understanding the constrained optimization problems inherent in financial engineering.

1.2 Statement of The Problem

The fundamental problem in algorithmic derivatives trading is the exponential complexity of the action space, which presents a significant combinatorial challenge for standard decision-making systems. In a typical trading environment, an agent must evaluate a deterministic chain of 80 options per timestep, encompassing two distinct expiries, twenty strike prices, and both call and put instrument types. When accounting for the various possible actions—buying, selling, or holding—and the selection of appropriate lot sizes for each, the number of potential multi-leg strategies becomes vast. Standard reinforcement learning architectures often struggle to navigate this discrete selection problem, as they lack the structural framework to generate coherent, multi-step trading sequences that effectively account for the interdependence of these variables.

Beyond the sheer volume of choices, the agent must manage a highly non-linear risk profile governed by complex financial sensitivities known as Greeks. Maintaining a delta-neutral state—where the aggregate portfolio sensitivity to changes in the underlying asset's price is kept near zero—is a continuous constrained optimization problem. This requires the precise calculation and tracking of Delta, Gamma, Theta, Vega, and Rho, all of which vary dynamically with market volatility and time to expiry. Without a sophisticated mechanism to prioritize these risk metrics, automated agents frequently converge on directional bets that offer high short-term returns but expose the portfolio to catastrophic losses during market shifts, fundamentally violating the core principles of risk-managed financial engineering.

Beyond the engineering challenge of navigating a high-dimensional action space, a deeper question motivates this research: can a transformer-based agent autonomously develop a structural understanding of the option chain — learning the difference between instrument types, expiry cycles, and moneyness regions — without any of these concepts being explicitly encoded? Existing reinforcement learning models are typically evaluated on output metrics alone, leaving the internal representations opaque. This project addresses both challenges simultaneously: developing an autoregressive Seq2Seq Transformer for delta-neutral trading and evaluating its learned representations through mechanistic interpretability of cross-attention weights.

1.3 Significance of The Study

The significance of this study lies in its novel application of Autoregressive Seq2Seq Transformers to the high-dimensional, combinatorial challenges of derivatives trading. By treating trade generation as a structured sequence—comprising actions, strikes, and lot sizes—this research demonstrates how modern neural architectures can navigate a vast search space involving a deterministic 80-option chain. The integration of cross-attention mechanisms allows for the simultaneously processing of multi-modal inputs, such as historical stock technicals and real-time theoretical Greeks, providing a more robust alternative to traditional feed-forward reinforcement learning models.

Furthermore, this work advances the field of risk-managed Reinforcement Learning by prioritizing portfolio stability over unconstrained profit maximization. Through the implementation of a Delta-Neutral Regularizer, the study provides a methodology for training agents to operate within strict risk parameters, such as maintaining a net delta threshold of 0.5. This approach is particularly significant for institutional trading environments where hedging and capital preservation are paramount, offering a scalable framework for solving the complex, non-linear optimization problems inherent in financial engineering.

Additionally, this study contributes to the emerging field of mechanistic interpretability in financial AI. By directly analyzing the cross-attention weights of the transformer decoder, the research provides a methodology for evaluating what financial concepts an RL agent has internalized — moving beyond black-box performance metrics toward transparent, interpretable evidence of learned financial structure. This is particularly significant for institutional applications where understanding why an agent makes a decision is as important as whether the decision is profitable.

1.4 Purpose of the Study

The primary objective of this study is to engineer a specialized Autoregressive Seq2Seq Transformer architecture that can effectively navigate the high-dimensional combinatorial landscape of derivatives trading. By representing the market state as a complex sequence of 111 tokens—incorporating 30 steps of historical stock technicals, a deterministic 80-option chain, and a "Rich Portfolio Summary"—the research aims to demonstrate that sequence-to-sequence learning can successfully map multi-modal financial inputs to structured, multi-leg trade orders. This approach shifts the paradigm from simple single-action selection to the generation of cohesive trading sequences that account for the deep interdependencies between different strikes, expiries, and instrument types.

Furthermore, this research establishes a methodology for using mechanistic interpretability to evaluate emergent financial understanding in RL agents. Rather than evaluating the agent solely on financial performance metrics, the study investigates whether the cross-attention mechanism autonomously develops representations of instrument type, expiry, and moneyness — concepts fundamental to options trading that were never explicitly provided as labels or rules. The delta-neutrality reward penalty serves as the training signal that induces this structural learning, rather than as the primary measure of success.

Ultimately, the study serves to evaluate the utility of cross-attention mechanisms in synthesizing disparate financial data points into actionable intelligence. By analyzing the attention masks, the research investigates the emergence of Instrument Type Awareness, Expiry Discrimination, and Moneyness Approximation as the agent learns the structural language of the option chain. This provides a scalable framework for financial engineering, offering a template for how modern AI architectures can be adapted to solve the complex constrained optimization problems inherent in institutional derivatives management.

1.5 Research Hypothesis

The central hypothesis of this research is that a reinforcement learning agent, through interaction with a custom trading environment, will autonomously develop a structural financial understanding of an option chain without explicit supervised encoding. This emergent understanding will manifest through three distinct mechanistic signatures in the transformer's cross-attention layers:

1. **Instrument Type Awareness:** The agent will learn the bilateral symmetry between Call (CE) and Put (PE) options, evidenced by the decoder attending to corresponding strikes across both instrument types to construct delta-neutral hedges.

2. **Expiry Discrimination:** The model will learn to distinguish between near and far expiry blocks, dynamically shifting attention between them as the temporal risks (Theta) and sensitivities (Gamma) accelerate closer to contract termination.

3. **Moneyness Approximation:** The cross-attention mechanism will autonomously track the "At-The-Money" (ATM) region, shifting its focus as the underlying asset price moves across the strike sequence, thereby performing implicit moneyness indexing.

By attending to the intersection of the underlying asset's current value and theoretical Greeks ($\Delta, \Gamma, \Theta, Vega, Rho$), the model is hypothesized to formulate complex, multi-leg strategies grounded in real-time risk sensitivities rather than simple price-action heuristics.

1.6 Definitions of Key Terms

To ensure clarity within this report, the following terms are defined as they relate to the implementation of the Autoregressive Trade Model and the Delta-Neutral trading environment.

Financial and Derivative Terms

Option (CE/PE): A derivative contract giving the buyer the right, but not the obligation, to buy (Call/CE) or sell (Put/PE) an underlying asset at a fixed strike price.

Strike Price (K): The pre-determined price at which an option holder can buy or sell the underlying asset upon exercise.

The Greeks: A set of risk measures that quantify the sensitivity of an option's price to various market parameters.

Delta (Δ): The rate of change of the option price relative to the price of the underlying asset; it is the primary metric for hedging.

Gamma (Γ): The rate of change of Delta relative to the underlying asset's price, representing the "acceleration" of the risk.

Theta (Θ): The rate of time decay, measuring the decrease in an option's value as it approaches its expiry date.

Vega: The sensitivity of the option's price to a 1% change in the implied volatility of the underlying asset.

Rho (Ρ): The sensitivity of the option's price to changes in the risk-free interest rate.

Option Chain: A deterministic listing of available option contracts for a specific asset, organized by expiry date, strike price, and type.

Moneyness: The relative relationship between the underlying price (S) and the strike price (K), specifically calculated in this project as S/K - 1.

Delta-Neutrality: A portfolio state where the aggregate Delta of all positions is zero or near zero, theoretically making the portfolio value immune to small price movements in the underlying asset.

Machine Learning and Architecture Terms

Reinforcement Learning (RL): A machine learning framework where an agent learns optimal behaviors by interacting with an environment to maximize a cumulative reward signal.

Proximal Policy Optimization (PPO): An on-policy RL algorithm that maintains training stability by clipping the policy update to prevent destructively large changes in the agent's behavior.

Transformer: A deep learning architecture that utilizes attention mechanisms to process sequences of data in parallel, effectively capturing long-range dependencies.

Sequence-to-Sequence (Seq2Seq): A model structure consisting of an Encoder to process input data (market states) and a Decoder to generate output sequences (trade orders).

Autoregressive Generation: A decoding process where the model predicts tokens one at a time, using its own previous outputs as context for the next prediction in the sequence.

Cross-Attention: A mechanism within the Transformer decoder that allows it to selectively focus on specific parts of the encoded market context, such as a particular strike in the option chain, when making a decision.

Reward Shaping: The inclusion of additional terms in the reward function—such as the Delta-Neutrality Penalty—to guide the agent toward specific risk-management objectives.

[CLS_PORT] Token: A specialized, learnable token prepended to the encoder sequence that acts as a summary representation of the agent's current portfolio state.

Experimental Context

Random Walk: A mathematical model of a series of random steps, used here to simulate underlying stock price movements where future prices are statistically independent of past trends.

LTP (Last Traded Price): The current market price of an option or stock as calculated via the Black-Scholes model or environmental update.

Lot Size: The standardized quantity of the underlying asset represented by one option contract, typically 50 in the context of the NIFTY50 index.

1.7 Research Limitations

A primary limitation of this study is the reliance on the Black-Scholes model for theoretical pricing and Greek calculations, which assumes constant volatility and log-normal return distributions that fail to account for the stochastic volatility and "fat tails" observed in real-world markets. Furthermore, while the Random Walk data generation provides a controlled environment for testing the agent's behavior, it lacks the structural complexities of live market regimes, such as liquidity constraints, institutional order flow, and "black swan" events that can decouple theoretical sensitivities from market reality. This gap between simulation and reality is further widened by the omission of transaction costs and slippage, which can significantly impact the net profitability of high-frequency options strategies. Additionally, the environment allows the agent to continue trading even when the portfolio value becomes negative, bypassing the liquidation events and margin calls that would typically terminate a live account.

The research is also constrained by the fixed architectural dimensions of the Autoregressive Trade Model, which utilizes a deterministic 80-option chain and a 30-step historical stock window. These static constraints may exclude relevant data from deep out-of-the-money strikes or long-term macroeconomic trends that influence derivative pricing over broader horizons. Similarly, the reward shaping parameters, specifically the 100.0 delta penalty and the 0.5 safety threshold, remain static throughout the training process. In a professional industry context, these parameters and the simplified 15% margin proxy would need to be dynamic to accommodate changing portfolio scales, fluctuating margin requirements, and complex risk-appetite regimes.

Chapter 2

This chapter establishes the dual pillars of the research: the intricate mechanics of derivative markets and the architectural advancements in deep learning that allow for complex sequence modeling. By first examining the fundamental principles of stock and option markets, we define the environment's constraints, such as the non-linear pricing behaviors dictated by the Black-Scholes model. This provides the necessary financial context to justify why high-dimensional selection problems—involving eighty distinct options across multiple strikes and expiries—require a departure from traditional linear models toward sophisticated neural architectures.

The second portion of the chapter details the transition from static trading rules to dynamic, agent-based learning through Reinforcement Learning (RL). We focus on the Sequence-to-Sequence (Seq2Seq) paradigm, which treats market observations as a complex input sequence to be mapped into an ordered output of trade actions. Central to this discussion is the Transformer's cross-attention mechanism, which provides the agent with the ability to selectively prioritize market technicals or specific option Greeks—such as delta, gamma, theta, vega, and rho—depending on the immediate portfolio risk-management objective.

Ultimately, these topics synthesize into a comprehensive framework for understanding how an autonomous agent can learn to maintain a delta-neutral state within a simulated environment. By combining Black-Scholes theoretical sensitivities with optimization algorithms like Proximal Policy Optimization (PPO), the study establishes a system capable of differentiating between market signal and noise. This theoretical review serves as the necessary prerequisite for the specific model architecture and experimental setup detailed in the following chapters.

2.1 Financial Derivatives and Trading

This section establishes the fundamental financial instruments and market mechanics that define the environment for the autonomous agent. By deconstructing the relationship between primary assets and their derived counterparts, we define the mathematical and operational constraints the agent must navigate to achieve delta-neutrality. This foundation is essential for understanding why a simple price-prediction model is insufficient and why a sophisticated sequence-to-sequence transformer is required to solve the complex structural selection problems inherent in modern derivatives management.

2.1.1 Overview of Stocks and Options

To understand the scope of this project, one must first distinguish between a primary asset and a derivative. A stock represents a direct share of ownership in a company or a market index, such as the NIFTY50. When owning a stock, the profit or loss profile is "linear": a one-unit increase in the stock price results in a corresponding one-unit gain for the investor, while a decrease results in an equivalent loss. This straightforward, one-to-one relationship makes stocks the "underlying" asset upon which more complex financial instruments are constructed.

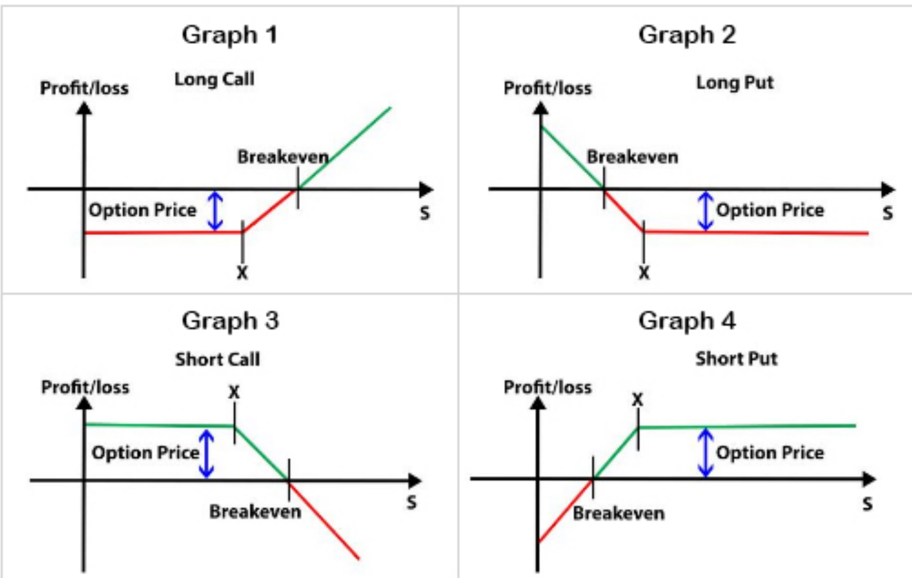

Figure 2.1 Comparison of Stocks and Options Payoff [1]

An option is a fundamentally different instrument known as a derivative, meaning its value is "derived" from the price of the underlying stock. Rather than owning the asset, an option is a contract that grants the trader the "right, but not the obligation" to buy or sell the stock at a specific price, known as the strike price, within a set timeframe. To acquire this right, the buyer pays an upfront fee called the premium, which represents the maximum possible loss for the buyer regardless of how far the market moves against them. Because options expire after a specific period, they are considered decaying assets; their value is determined not only by the stock price but also by the time remaining before the contract reaches its expiry date.

In this simulation, the agent interacts with two primary types of options: Calls (CE) and Puts (PE). A Call option gives the trader the right to buy the stock at the strike price, essentially acting as a bet that the market will rise. Conversely, a Put option provides the right to sell the stock, functioning as a bet that the price will fall or as insurance against market downturns.

To clarify these mechanics, consider a practical example involving the NIFTY50 index. Assume the index is currently trading at an underlying price of 18,000. A trader buys a Call option with a strike price of 18,100, paying a premium (option price) of 100. This contract allows the trader to buy the index at 18,100, no matter how high it climbs. On the day of expiry:

If the price rises to 18,500: The option is "in-the-money." The gross profit is the difference between the market price and the strike (18,500 - 18,100 = 400). After subtracting the initial premium of 100, the net profit is 300.

If the price falls to 17,900: It would not be rational to buy the index at 18,100 when it is available in the market for less. The option expires worthless, and the trader's net loss is 100, capped at the premium paid.

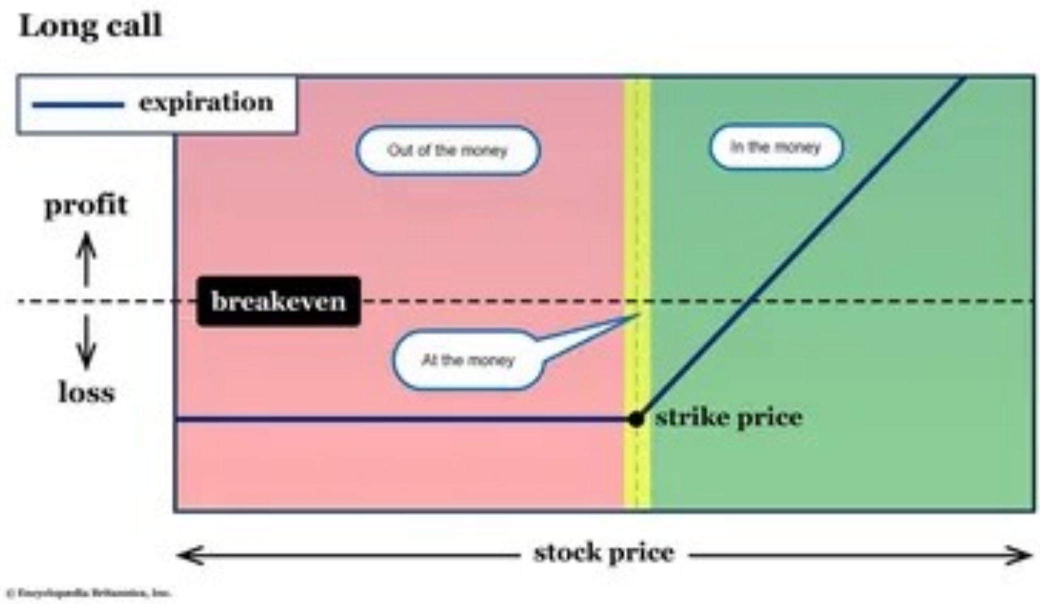

Figure 2.2 Comparison of Options Payoff [5]

To facilitate a stable training environment, this project utilizes a deterministic option chain consisting of exactly 80 options per timestep. This fixed chain is composed of 40 Call options and 40 Put options, distributed across two expiry dates. Specifically, each of the two expiries contains 20 strikes, with each strike having one Call and one Put contract. This standardized structure allows the transformer model to process a constant input size of 80 tokens, ensuring consistency during the encoding and decoding phases.

While this project uses a fixed 80-option chain to ensure architectural consistency, in a real-world market, the size and structure of an option chain are dynamic. As seen in the provided screenshot of the NSE NIFTY 50 Option Chain, the live market presents dozens of available strike prices that expand as the index price moves.

The NSE table is organized with Calls on the left and Puts on the right, with the Strike Price column acting as the central axis. The table is for underlying stock index price being 24,378 INR. Key data points in this table include:

CALLS											PUTS									
OI	CHNG IN OI	VOLUME	IV	LTP	CHNG	BID QTY	BID	ASK	ASK QTY	STRIKE	BID QTY	BID	ASK	ASK QTY	CHNG	LTP	IV	VOLUME	CHNG IN OI	OI
455	-10	137	-	764.10	-206.10	65	765.55	772.35	260	23,650.00	65	31.15	31.40	1,040	-3.25	31.50	20.03	1,40,938	4,023	8,315
3,463	-290	2,339	13.17	723.30	-197.65	520	721.05	726.30	260	23,700.00	845	36.00	36.30	2,340	-2.30	36.30	19.86	3,13,181	4,246	27,323
577	-54	226	14.38	679.95	-196.20	130	676.35	682.20	260	23,750.00	715	41.40	41.80	715	-1.15	41.45	19.65	1,83,224	2,673	8,752
7,154	-321	5,140	14.34	633.60	-196.60	260	633.85	636.00	65	23,800.00	3,185	48.00	48.05	325	0.25	48.00	19.54	4,18,822	-2,235	35,428
1,434	61	684	14.34	588.30	-197.15	585	592.50	596.25	260	23,850.00	1,365	55.00	55.35	715	1.75	54.90	19.36	1,78,052	1,969	7,504
4,941	-139	6,777	15.26	550.00	-190.90	325	550.95	553.80	195	23,900.00	195	63.25	63.55	195	4.25	63.25	19.25	4,15,276	11,945	44,735
934	196	2,109	14.95	505.25	-194.00	325	510.55	512.75	195	23,950.00	325	72.70	73.00	260	6.10	72.70	19.15	2,70,253	3,621	7,826
48,108	-3,685	89,274	15.85	471.55	-183.70	325	470.75	472.40	195	24,000.00	715	83.00	83.35	195	9.35	83.00	19.02	13,12,700	5,521	1,19,285
896	130	5,528	16.04	434.15	-180.45	325	433.45	434.95	195	24,050.00	195	94.90	95.00	130	13.45	94.95	18.93	3,19,169	4,788	8,973
9,033	437	30,461	16.05	396.70	-177.60	325	396.75	397.70	195	24,100.00	195	108.30	108.70	390	18.20	108.70	18.89	5,57,896	9,224	31,191
1,591	307	16,304	16.17	362.20	-172.80	325	361.65	362.60	195	24,150.00	195	122.75	123.30	195	22.05	123.25	18.80	2,77,661	1,909	9,117
19,269	1,804	1,28,459	16.14	327.80	-168.60	260	326.55	327.90	195	24,200.00	390	138.85	138.95	195	26.95	138.85	18.66	7,00,859	2,014	38,969
5,575	1,704	76,590	16.22	296.60	-162.10	585	296.15	296.95	195	24,250.00	455	156.70	157.25	260	32.70	157.15	18.63	4,38,868	5,085	11,490
38,144	16,302	4,42,034	16.09	264.65	-158.30	65	265.00	265.50	195	24,300.00	130	176.40	177.10	260	38.45	176.40	18.54	10,41,118	19,845	54,436
13,529	9,978	4,51,948	16.14	236.85	-150.50	325	237.15	237.80	455	24,350.00	130	198.00	198.30	260	46.10	198.30	18.53	8,40,904	12,416	22,616
81,587	51,804	19,12,096	16.10	209.85	-143.70	455	209.90	210.65	455	24,400.00	520	220.75	221.60	195	52.75	221.70	18.51	21,47,903	31,312	78,029
36,908	32,107	12,45,501	16.15	186.00	-135.45	260	185.30	186.00	65	24,450.00	715	245.35	246.20	65	60.80	246.20	18.43	10,62,088	11,294	20,945
1,19,938	48,037	18,77,282	16.04	162.15	-128.15	1,105	162.15	162.75	195	24,500.00	130	272.10	272.95	65	68.45	272.95	18.40	12,23,572	-14,070	72,617
24,225	8,512	5,02,280	16.08	142.10	-120.15	2,405	142.00	142.35	455	24,550.00	195	301.45	302.40	195	77.85	301.40	18.37	2,02,366	-7,514	12,567
56,547	22,906	8,13,991	16.09	123.45	-110.85	715	123.00	123.10	130	24,600.00	1,755	331.80	332.90	195	86.55	333.00	18.46	2,58,478	-8,259	19,738
14,958	8,437	3,32,159	16.03	105.95	-102.65	65	105.50	105.90	195	24,650.00	975	364.40	365.80	195	96.80	368.50	18.75	40,023	-44	2,958
41,870	13,416	6,51,105	16.02	90.75	-93.90	390	90.40	90.65	195	24,700.00	195	399.10	400.40	195	104.45	401.00	18.65	74,094	228	8,909
16,160	11,160	3,65,749	15.98	77.05	-84.75	195	77.05	77.35	195	24,750.00	195	436.15	438.05	260	118.70	443.20	19.31	12,790	82	891
56,711	28,172	7,12,481	15.99	65.30	-75.45	910	65.10	65.30	520	24,800.00	195	473.75	475.25	195	124.90	479.60	19.33	37,872	-651	10,412
15,392	10,805	2,55,137	16.02	55.20	-66.85	1,040	55.00	55.15	1,105	24,850.00	260	513.25	515.35	195	133.25	519.80	19.61	5,292	339	774
55,920	28,556	5,11,388	15.97	45.75	-59.10	130	45.75	45.90	975	24,900.00	260	553.00	557.35	260	142.80	560.60	19.84	9,973	-258	2,396
11,570	7,561	2,11,183	16.04	38.50	-50.80	65	38.20	38.50	1,430	24,950.00	260	595.80	598.80	325	148.20	603.15	20.17	1,376	163	421
1,53,683	49,930	9,92,533	16.11	32.30	-43.20	2,015	31.90	32.30	1,235	25,000.00	130	640.00	641.85	260	153.15	640.40	19.77	24,117	-351	32,057
14,565	8,448	1,92,281	16.11	26.55	-37.20	975	26.20	26.55	975	25,050.00	260	683.75	690.55	260	168.45	694.70	21.36	654	162	270
35,079	6,710	3,37,417	16.20	22.10	-31.15	1,235	21.70	21.95	520	25,100.00	260	729.15	735.85	260	175.90	740.00	21.82	1,503	126	800

Figure 2.3 Snapshot of NIFTY 50 Full Option Chain [4]

LTP (Last Traded Price): The current premium or cost of the option.

IV (Implied Volatility): A measure of the market's expectation of future price swings.

OI (Open Interest) and Volume: Indicators of liquidity and the number of active contracts.

Shaded Regions: The yellow-shaded areas indicate "In-the-Money" (ITM) options, which possess intrinsic value because the strike price is favorable compared to the current index level.

The complexity for the agent arises from the non-linear nature of option pricing, often modeled by the Black-Scholes framework. Unlike a stock, an option’s price does not move in a simple ratio with the underlying asset. This sensitivity is quantified by the Greeks—Delta, Gamma, Theta, Vega, and Rho—which provide the agent with the data needed to understand how portfolio value will fluctuate based on market moves, time decay, and changes in volatility. The ultimate goal of Delta-Neutrality is to use these 40 Calls and 40 Puts to cancel out directional risks, creating a portfolio that is immunized against small fluctuations in the underlying asset's price.

2.1.2 Mechanics of Derivatives Trading

The core objective of derivatives trading is often not pure speculation, but hedging—a risk-management strategy used to offset potential losses in one investment by making another. In this project, the most critical form of hedging is Delta-Neutral Hedging. To understand this, one must first understand Delta (Δ), which is the "sensitivity" of an option's price to the underlying stock's movement. If an option has a Delta of 0.6, it means that for every $1 move in the stock price, the option's value is expected to change by approximately $0.60. Delta acts as a multiplier that tells the trader—and our autonomous agent—how much directional risk they are carrying at any given moment.

A Delta-Neutral state is achieved when the total Delta of all positions in a portfolio adds up to zero. In this state, the portfolio is "neutralized" against small price swings in the underlying asset; whether the market goes up slightly or down slightly, the total value of the portfolio remains relatively stable. To reach this balance, traders use a combination of multiple options to cancel out each other's sensitivities. For example, if a trader holds a position that gains money when the market rises (Positive Delta), they must balance it with a position that gains money when the market falls (Negative Delta).

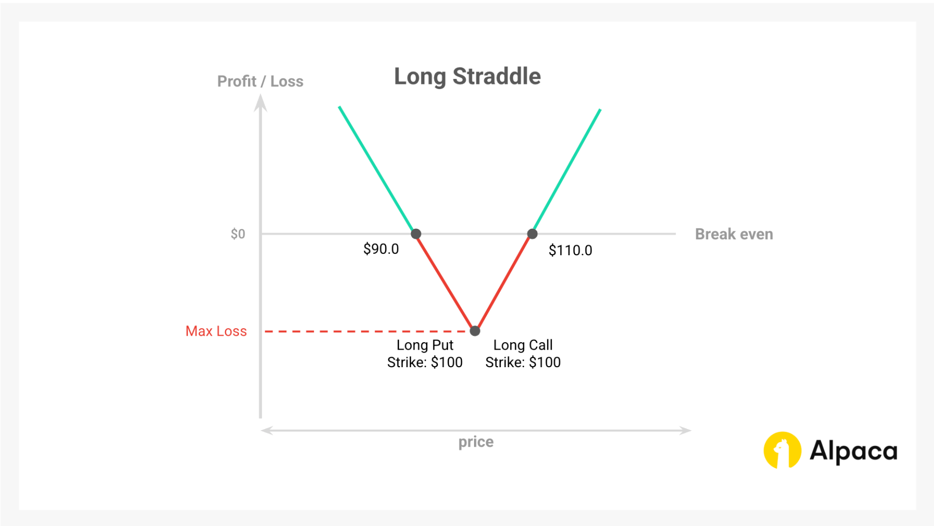

Figure 2.4 Long Straddle Hedging Strategy Payoff [2]

The fundamental mechanics of call options are defined by a linear but asymmetrical risk-reward profile. As illustrated in the provided figure, the Long Call position (top) represents a "right but not an obligation" to purchase the underlying asset. The holder's risk is strictly limited to the premium paid (C_0), while the potential for profit is theoretically unlimited as the asset price S rises above the strike price K. The break-even point occurs at $S = K + C_0$. Conversely, the Short Call position (bottom) represents the obligation to sell the asset if the option is exercised. The seller receives the premium (C_0) as a maximum fixed gain but faces theoretically unlimited risk should the asset price significantly appreciate. This mirror-image relationship underscores the zero-sum dynamics of the derivatives market and highlights the intrinsic value calculation:

Long Call P/L $= \max(0, S - K) - C_0$

Short Call P/L $= C_0 - \max(0, S - K)$

In the development of the Autoregressive Trade Model (ATR), these payoffs serve as the ground truth for the environment's reward function. While the agent aims for delta-neutrality, it must navigate these underlying payoff structures to decide which strikes (K) to enter or exit based on the current market price S and the Greeks derived from these very curves.

To illustrate this with a calculation, imagine the NIFTY50 index is at 18,000. An agent buys one Call option that has a Delta of 0.5. At this point, the portfolio's total Delta is +0.5. If the index moves up by 10 points to 18,010, the portfolio value will rise by 5 points (10 x 0.5). To become delta-neutral, the agent could simultaneously sell (short) a different Call option with a Delta of 0.5, or buy a Put option with a Delta of -0.5.

Initial Position: 1 Call (Delta = +0.5)

Hedge Position: 1 Put (Delta = -0.5)

Total Portfolio Delta: $0.5 + (-0.5) = 0

Now, if the market moves up 10 points, the Call gains 5 points, but the Put loses 5 points. The net change is zero. The agent has successfully "insured" the portfolio against directional movement, allowing it to remain stable regardless of small market fluctuations.

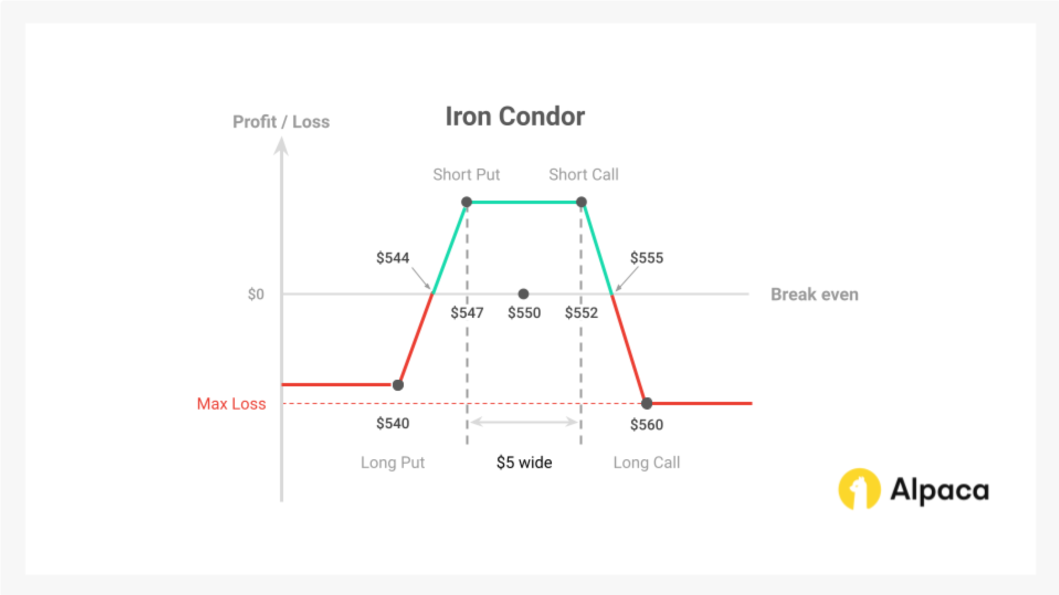

Figure 2.5 Iron Condor Hedging Strategy Payoff [3]

In professional trading, this balancing act is often structured into specific strategies such as Spreads or Iron Condors. A Call Spread, for instance, involves buying one Call and selling another at a different strike price to limit both the cost and the risk of the trade. An Iron Condor is a more advanced delta-neutral strategy where the trader sells both a Call Spread and a Put Spread simultaneously. This creates a "profit zone" where the trader earns money as long as the market stays within a certain range, effectively betting on stability rather than direction. These strategies are the building blocks the agent uses to navigate the 80-option chain provided in the environment.

While delta-neutrality is the primary goal, we are teaching our reinforcement learning agent to manage this state while considering other "Greeks" that add layers of complexity. The agent must account for Gamma, which measures how quickly the Delta itself changes as the market moves, and Theta, the relentless decay of value over time. If the agent ignores Gamma, a delta-neutral portfolio can quickly become "unbalanced" during a sudden market spike. By using a specialized reward function that applies a heavy penalty when the absolute Delta exceeds 0.5, the training process forces the agent to constantly "juggling" these multiple sensitivities. The result is a highly sophisticated system that uses the cross-attention mechanism to weigh price signals against time decay, ensuring a stable and risk-aware trading policy.

2.2 Deep Learning Algorithms

2.2.1 Deep Learning Principles

At the core of this study's architectural framework are the principles of deep learning, specifically the capacity of artificial neural networks to act as universal function approximators. These systems are designed to map complex, high-dimensional input spaces—such as the 111-token market state used in this project—to specific target outputs through successive layers of mathematical abstraction. This mapping is achieved by optimizing a set of internal weights and biases to minimize a loss function, $J(\theta)$, which quantifies the error between the model's current output and the desired target. Through the process of backpropagation and gradient descent, the network iteratively refines its parameters by calculating the gradient of the loss with respect to each weight, allowing the model to "learn" the subtle, non-linear relationships inherent in financial derivatives that would be impossible to capture with traditional linear regression.

Beyond simple input-output mapping, deep learning emphasizes representational learning, where the model autonomously identifies the most relevant features within raw data. By utilizing non-linear activation functions, the network can model the complex "hockey-stick" payoff structures and volatility smiles that define the options market. These foundational concepts of feature extraction and iterative optimization provide the necessary bedrock for the more advanced components of this research. For instance, the transition from basic multilayer perceptrons to sequence-aware models allows for the conversion of discrete option strikes into continuous embeddings, while the optimization framework is adapted into Proximal Policy Optimization (PPO) to refine the agent's strategy based on a reward signal rather than a static label. This functional flexibility is what enables the autoregressive agent to synthesize historical technicals with instantaneous Greek sensitivities to maintain a delta-neutral portfolio.

2.2.2 Reinforcement Learning (RL) for Trading

Reinforcement Learning (RL) represents a paradigm shift from traditional supervised learning by focusing on goal-oriented behavior through interaction rather than static labels. In this framework, an autonomous agent learns to make a sequence of decisions by interacting with an uncertain, stochastic environment—in this case, the simulated NIFTY50 options market. The process is defined as a continuous loop where, at each timestep t, the agent observes the current state (S_t), executes an action (A_t), and receives a numerical reward (R_{t+1}) along with the subsequent state (S_{t+1}). The ultimate objective of the agent is to learn an optimal policy (π), which is a mapping from states to actions that maximizes the expected cumulative reward over time. This trial-and-error approach is particularly suited for derivatives trading, where the "correct" trade is not always obvious and must be discovered by evaluating the long-term impact of risk sensitivities and capital allocation.

In the specific context of this study, the RL framework is adapted to handle a high-dimensional, combinatorial action space through Proximal Policy Optimization (PPO). Standard RL algorithms often suffer from instability in financial environments due to high noise and the "policy collapse" that occurs when an agent makes excessively large updates based on temporary market anomalies. PPO mitigates this by using a specialized objective function that "clips" the policy update, ensuring the new strategy does not deviate too drastically from the previous one. Furthermore, the traditional pursuit of profit is constrained through reward shaping, where the agent's feedback is not merely its Profit and Loss (PnL), but a composite score that penalizes the absolute value of the portfolio's

Delta. By applying a heavy penalty when the aggregate Delta exceeds the 0.5 safety threshold, the RL framework forces the agent to internalize the principles of delta-neutral hedging, transforming the optimization problem into a sophisticated balancing act of risk and reward.

2.2.3 Transformer and Cross-Attention Mechanisms

The Transformer architecture represents a significant departure from traditional recurrent neural networks (RNNs) by utilizing attention mechanisms to process entire sequences of data in parallel rather than sequentially. This parallelization is achieved through self-attention, which allows each token in a sequence—such as a specific strike price in the 80-option chain—to interact with every other token regardless of their distance. In the context of this study, the Encoder transforms the 111-token market state into a high-dimensional latent representation, capturing the intricate relationships between historical price trends and the instantaneous sensitivities of the derivatives.

The critical bridge between the market context and the agent's decision-making is the Cross-Attention mechanism. Unlike self-attention, cross-attention allows the Decoder (which generates the trading actions) to selectively "query" the Encoder's memory of the market. This is mathematically modeled through a Query-Key-Value (QKV) operation: the decoder generates a Query (Q) based on its current action sequence, while the encoder provides Keys (K) and Values (V) derived from the market state. By calculating the dot product between the Query and the Keys, the model produces attention weights that determine which market features—such as an At-The-Money strike or a high-volatility technical indicator—should most influence the current trade order.

This mechanism is what enables the agent to approximate option moneyness and exhibit "rational temporal ignorance". Because the cross-attention layers can dynamically reweight the 111 input tokens at every decoding step, the agent can learn to ignore historical "noise" when the underlying asset follows a random walk, focusing instead on the instantaneous portfolio summary to maintain delta-neutrality. This flexibility allows the AutoregressiveTradeModel to formulate complex, multi-leg strategies like Iron Condors by focusing specifically on the relevant strike prices in the encoder’s memory while generating each triplet of the action sequence.

2.2.4 Sequence to Sequence (Seq2Seq) Models

Sequence-to-Sequence (Seq2Seq) models are a class of deep learning architectures designed to map a source input sequence to a target output sequence, even when the two have different lengths or structural formats. Traditionally, this framework is composed of two primary components: an Encoder and a Decoder. The Encoder processes the entire source sequence to produce a dense latent representation—a "context vector"—that encapsulates the most salient information from the input. The Decoder then takes this representation and generates the target sequence token-by-token. This paradigm is particularly effective for complex translation tasks, such as converting a high-dimensional market state into a structured series of executable trade orders.

In this project, the Seq2Seq paradigm is implemented through an Autoregressive Trade Model. The input is a 111-token sequence representing the market environment—including historical stock technicals and the 80-option chain—while the output is a variable-length sequence of trade orders structured as triplets: Action, Strike, and Lot Size. By framing trading as a sequence generation task, the model can generate cohesive multi-leg strategies. For instance, when constructing a delta-neutral position, the choice of a specific strike price and its corresponding lot size is directly conditioned on the actions previously generated in that sequence.

The autoregressive nature of the decoder is the engine behind this sequential logic. During generation, the token predicted at step k is fed back into the decoder as part of the context for step k+1. This feedback loop ensures that the agent "remembers" the legs of the trade it has already committed to within the current timestep, allowing it to dynamically balance its choices. This process mimics the workflow of a professional derivatives trader who builds a complex portfolio leg-by-leg, ensuring that each new position serves to offset the risk of the previous ones to maintain an aggregate delta-neutral state.

Chapter 3

3.1 Model Architecture

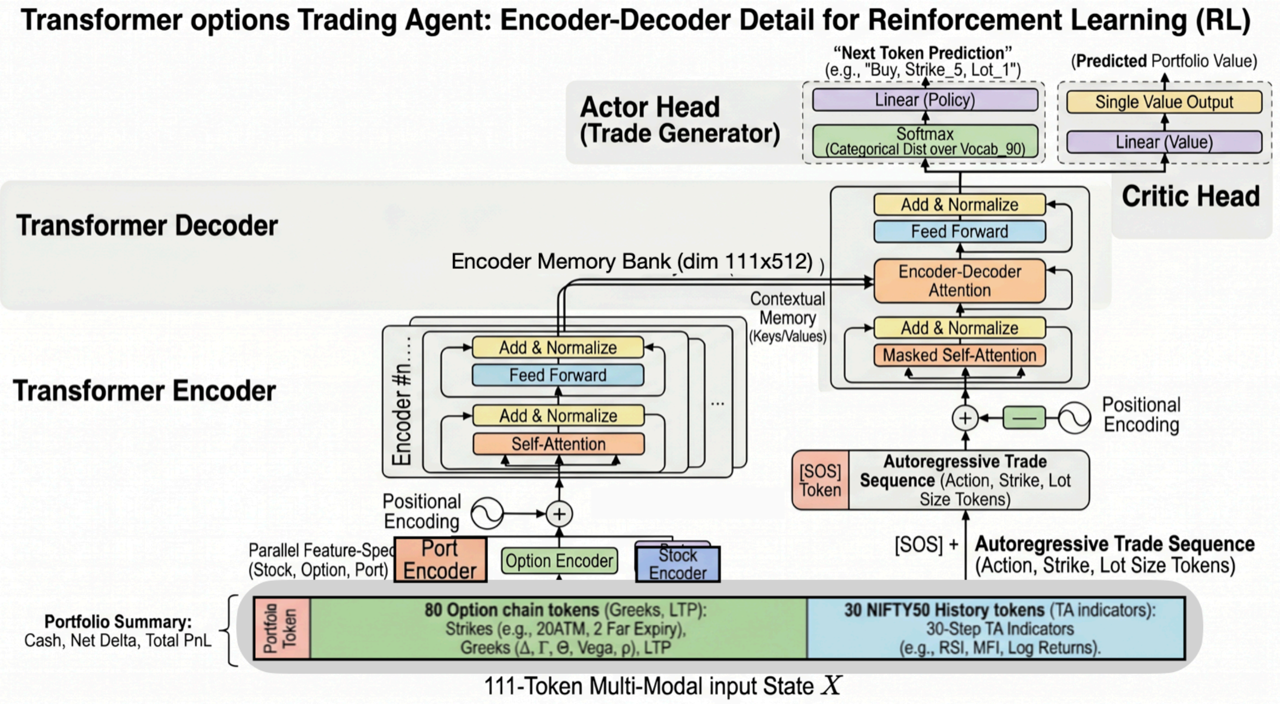

Figure 3.1 Model Architecture

The model architecture, titled Transformer Options Trading Agent, is a sophisticated Sequence-to-Sequence (Seq2Seq) Transformer specifically engineered for high-dimensional financial environments. The process begins at the 111-Token Multi-Modal Input State, where three disparate data streams are fused into a unified dense representation. This input includes a specialized [CLS_PORT] Portfolio Token (tracking cash, net delta, and PnL), 80 Option Tokens (capturing the full chain's Greeks, strikes, and pricing), and 30 Stock History Tokens (composed of technical indicators like RSI and MFI). Rather than a single projection, the model utilizes Parallel Feature-Specific Encoders to map each data type into a shared 512-dimensional latent space. This ensures that the distinct mathematical properties of historical stock trends and instantaneous option Greeks are preserved before they are combined for processing.

At the core of the agent is a dual-stack Transformer framework designed for Mechanistic Interpretability. The Transformer Encoder processes the 111-token state through multiple layers of self-attention to generate a Contextual Memory Bank, which serves as an internal representation of market risk and opportunity. The Transformer Decoder then operates autoregressively to construct trades token-by-token, beginning with a Start of Sequence [SOS] token. The critical mechanism here is the Encoder-Decoder Attention (Cross-Attention), where the decoder "queries" the encoder's memory to ground its trade generation in current market realities. To prevent the model from "looking ahead" during training, the decoder utilizes Masked Self-Attention, ensuring each generated token—Action, Strike, and Lot Size—is conditioned only on previous decisions and the market state.

The final layer of the architecture splits into a dual-head configuration to facilitate Reinforcement Learning (RL) via the Proximal Policy Optimization (PPO) algorithm. The Actor Head (Trade Generator) applies a linear policy layer and a softmax function to produce a probability distribution over a 90-Token Vocabulary, which is then used for Next Token Prediction. Simultaneously, the Critic Head (Value Function) outputs a single scalar value representing the Predicted Portfolio Value, providing the baseline used to calculate the advantage function during training. A unique Feedback Loop (Feed to ATR) ensures that each predicted token is fed back into the decoder, allowing the agent to construct complex, multi-part trades that are structurally consistent with its objective of maintaining delta-neutrality while optimizing for PnL.

3.1.1 Sequence to Sequence Learning Architecture

The core of the system is a Sequence-to-Sequence (Seq2Seq) architecture that functions as a sophisticated "translator," mapping a 111-token market state to a variable-length sequence of trade actions. Unlike traditional Reinforcement Learning models that output a single probability distribution over a fixed action space, this Seq2Seq approach allows the agent to generate multi-leg strategies by treating each trade as a structured sentence. The model is composed of a Transformer Encoder and a Transformer Decoder, linked by a Cross-Attention mechanism that ensures every trade decision is grounded in both the current market reality and the agent's internal portfolio risk.

The Encoder serves as the sensory organ of the agent, processing a high-dimensional input sequence consisting of 111 specialized tokens. These include:

- 80 Option Tokens: Representing the Greeks (Delta, Gamma, Theta, Vega, Rho) and pricing data for a deterministic 80-option chain.
- 30 Historical Stock Tokens: Capturing a 30-step window of technical indicators like RSI and MFI.
- 1 [CLS_PORT] Token: A specialized summary token representing the agent's current portfolio state, including cash balance and net delta exposure.

The Decoder then generates the trade orders using Autoregressive Decoding. It produces tokens in triplets—Action (Buy/Sell/Hold), Strike, and Lot Size—one at a time. Because it is autoregressive, each subsequent token is conditioned on the tokens previously generated within the same timestep. This allows the agent to maintain internal consistency; for example, if the decoder generates a "Sell" action for a specific strike, the next "Lot Size" token is selected based on the realization that it is filling a short position, ensuring the generated sequence forms a coherent financial strategy.

3.1.2 Transformers Integration

The integration of the Transformer architecture provides the computational framework for the agent's policy network, π_θ, enabling the simultaneous processing of multi-modal market data. By utilizing multi-head self-attention, the encoder produces contextualized embeddings for the 111-token input sequence, which captures the spatial relationships between the 80 option strikes and the temporal dependencies of the 30-step historical stock technicals. This high-dimensional latent representation ensures that each token is encoded with global context, allowing the model to distinguish between relevant market signals and stochastic noise before passing information to the decoder.

The architectural bridge between the encoded market state and trade generation is the cross-attention mechanism. Within this integration, the decoder generates queries derived from the partially generated action sequence, which are matched against the keys and values provided by the encoder's memory bank. This specific operation allows the model to dynamically weight theoretical Greeks and technical indicators based on the immediate needs of the portfolio. By calculating attention scores across the option chain, the agent identifies At-The-Money (ATM) regions and relevant hedging instruments without requiring hard-coded rules for moneyness indexing.

This Transformer-based policy is integrated directly into the Proximal Policy Optimization (PPO) framework to ensure stable convergence within the complex action space. The autoregressive nature of the decoder facilitates the production of structured action triplets—Action, Strike, and Lot Size—where each decision is conditioned on the preceding tokens in the sequence. During training, the PPO algorithm optimizes the Transformer's internal weights based on a shaped reward function that enforces delta-neutrality. This integration allows the agent to iteratively refine its attention masks, eventually learning to prioritize the portfolio summary and current Greek sensitivities to maintain the aggregate delta within the 0.5 safety threshold.

3.1.3 Implementation of Cross-Attention Mechanism

The implementation of cross-attention within the Autoregressive Trade Model functions as the primary interface between the encoded market environment and the sequence of trade actions. In each decoder block, the model executes a multi-head cross-attention operation where Queries (Q) are derived from the partially generated action sequence, and Keys (K) and Values (V) are provided by the encoder's context-aware memory of the 111-token market state. The attention mechanism computes the alignment between these components using the scaled dot-product formula:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

This mathematical operation produces a dynamic weighting across the market state, which is the mechanism used to evaluate the hypothesis of emergent structural awareness (Instrument, Expiry, and Moneyness).

This mechanism is the driver for the agent's ability to approximate moneyness and maintain delta-neutrality without hard-coded indexing. Because the attention weights are re-calculated for every token in the action triplet (Action, Strike, Lot Size), the agent can "query" the 80-option chain for specific Greeks that offset its current risk exposure. For instance, if the [CLS_PORT] token indicates a net positive delta, the cross-attention weights typically shift toward the Put strikes or the "Sell" side of the Call chain to identify hedging instruments. This ability to perform a high-speed, latent lookup across the 111 input tokens ensures that each generated trade is grounded in the theoretical sensitivities (Delta, Gamma, Theta, Vega, Rho) necessary for capital preservation.

Furthermore, the implementation of cross-attention facilitates the agent's "rational temporal ignorance" within the simulation. By analyzing the relationship between the input tokens and the cumulative reward signal, the attention mechanism autonomously identifies that historical stock tokens (81–110) are less predictive in a random walk environment than the instantaneous Greeks (0–80). Consequently, the decoder learns to minimize the attention weights assigned to the 30-step history, effectively pruning "noise" from its decision-making process. This focused allocation of attention ensures that the policy remains robust and computationally efficient, prioritizing the immediate delta-neutrality constraint over statistically irrelevant historical trends.

3.1.4 Proposed Model

The proposed architecture, designated as the Autoregressive Trade Model (ATR), is a specialized Transformer-based system engineered to map a high-dimensional market state to a structured sequence of executable trade orders. Unlike traditional reinforcement learning models that utilize a flat action vector, the ATR model treats trade generation as a sequence-to-sequence translation task. This design allows the agent to synthesize a multi-modal input—comprised of a 30-step historical stock sequence, a deterministic 80-option chain, and a real-time portfolio summary—into a cohesive multi-leg strategy. By leveraging the parallel processing of the Transformer encoder and the sequential logic of the autoregressive decoder, the model provides a scalable solution for navigating the combinatorial complexities of the NIFTY50 options market.

To process disparate financial data types within a unified framework, the ATR model utilizes specialized linear projection layers that map the 111 input tokens into a common 512-dimensional latent space. The encoder processes these tokens—specifically the [CLS_PORT] portfolio summary, the 80 option Greeks, and the 30-step stock history—to produce a context-aware representation of the environment. Multi-head self-attention, implemented with 4 attention heads, allows the model to capture deep interdependencies between different strikes and historical technical indicators. This unified embedding approach ensures that the subsequent trade decisions are grounded in both external market regimes and the portfolio's internal risk sensitivities.

The decoder generates trade orders through an autoregressive process, producing action triplets consisting of Action (Buy, Sell, or Hold), Strike Price, and Lot Size. Each token in the triplet is conditioned on the preceding tokens in the sequence, ensuring that the generated trade legs are financially consistent and logically structured. This generation process is optimized via the Proximal Policy Optimization (PPO) algorithm, utilizing a reward function that enforces delta-neutrality through a heavy penalty ($\lambda = 100.0$) when the portfolio's absolute delta exceeds the 0.5

threshold. By integrating sequence-to-sequence learning with risk-aware reinforcement learning, the ATR model autonomously learns to prioritize portfolio stability over unconstrained speculation, effectively using its cross-attention mechanism to identify optimal hedging instruments within the 80-option chain.

3.2 Data Engineering

3.2.1 Data Sourcing: Stocks and Options

The underlying stock data is generated through a geometric Brownian motion process to simulate the stochastic price movements of the NIFTY50 index. This simulation produces a continuous time series of prices, which are utilized to calculate technical indicators including the Relative Strength Index (RSI) and the Money Flow Index (MFI). These indicators provide a 30-step historical context for the model's encoder. The use of a random walk ensures that the environment is Markovian, where future price movements are statistically independent of historical trends. This allows for the empirical evaluation of the agent's "rational temporal ignorance," specifically its ability to prioritize instantaneous market data over non-predictive historical sequences.

Option data is derived from the simulated stock prices using the Black-Scholes pricing model. For each timestep, the system calculates the theoretical price and the corresponding Greeks—Delta, Gamma, Theta, Vega, and Rho—for every contract in the sequence. The calculation assumes fixed parameters, including a risk-free interest rate of 5% and an annual volatility of 20%. This synthetic generation provides a high-fidelity dataset that captures the non-linear relationship between the underlying asset price and derivative values, providing the multi-modal input necessary for the agent to learn delta-neutral hedging behaviors.

To ensure architectural consistency for the Transformer model, the option data is organized into a deterministic chain of 80 contracts. This chain consists of 40 Call options and 40 Put options, divided equally between near-term and far-term expiry cycles. For each expiry, the system selects

20 strike prices centered around the current At-The-Money (ATM) price. This structured representation ensures that the encoder receives a fixed-length input of 80 option tokens per timestep, providing the data necessary for the agent to autonomously learn structural signatures like instrument symmetry and moneyness.

3.2.2 Synthetic Data: Random Walk Data Generation

The underlying asset price for the simulation is generated using a geometric Brownian motion (GBM) process to approximate the stochastic price behavior of the NIFTY50 index. This random walk generation ensures that the environment is strictly Markovian, meaning the probability distribution of future prices depends only on the current price rather than the historical sequence. By utilizing a controlled synthetic environment where historical trends have no predictive power, the research can empirically evaluate whether the agent learns to prioritize instantaneous Greeks over historical stock data, a behavior defined as "rational temporal ignorance".

The generation process produces a continuous time series from which a 30-step historical window is sampled at each timestep. From this raw price data, two primary technical indicators are calculated: the Relative Strength Index (RSI) and the Money Flow Index (MFI). These indicators, along with the log returns of the price series, are normalized and projected into the latent space as the 30 historical stock tokens. This setup provides the multi-modal input necessary for the encoder to process temporal patterns, even if those patterns are statistically irrelevant for future price prediction within the random walk framework.

3.2.3 Black-Scholes Option Data Modeling

The theoretical valuation of the 80-option chain is performed using the Black-Scholes-Merton model for European-style options. This model provides the mathematical framework to calculate the Last Traded Price (LTP) and the corresponding Greeks based on five primary inputs: the underlying asset price (S), the strike price (K), the time to expiration (T), the risk-free interest rate (r), and the asset's annualized volatility (σ). In the simulation environment, the risk-free rate is set to a constant 5% (r=0.05) and the volatility is fixed at 20% ($\sigma = 0.20$) to maintain a controlled data distribution for reinforcement learning.

Each option token within the 111-token market state includes five theoretical Greeks: Delta (Δ), Gamma (Γ), Theta (Θ), Vega, and Rho (ρ). These sensitivities are calculated by taking the partial derivatives of the Black-Scholes pricing formula with respect to the underlying parameters. Delta represents the rate of change of the option price relative to the underlying price and serves as the primary metric for the agent's hedging objective. Gamma measures the rate of change in Delta, while Theta quantifies time decay, Vega tracks sensitivity to volatility, and Rho measures sensitivity to interest rate changes. By providing these five Greeks as features for each of the 80 option tokens, the encoder captures the non-linear risk profile of the entire option chain.

The modeling process ensures that the option chain remains deterministic and centered around the current market price. At each timestep, the system selects 20 strike prices per expiry cycle—covering both near-term and far-term expiries—with 10 strikes positioned above the At-The-Money (ATM) price and 10 strikes below. For every strike, both a Call and a Put option are modeled, resulting in 40 options per expiry and 80 options total. This structured data modeling allows the transformer encoder to process a consistent input space, facilitating the cross-attention mechanism's ability to map instantaneous market sensitivities to trade generation.

3.2.4 Options Expiry and Relative Data Generation

The data generation process incorporates two distinct expiry cycles to model the temporal risks inherent in derivatives trading: the "Near Expiry" and the "Far Expiry". In the simulation environment, the Near Expiry represents the current month's contract cycle, while the Far Expiry represents the subsequent month. This dual-expiry structure is critical for training the agent to manage time decay (Theta) and roll positions across different temporal horizons. As time T approaches zero for the Near Expiry, the sensitivity of the option price to the underlying movement increases, necessitating active rebalancing by the agent to maintain a delta-neutral state.

To maintain a consistent input representation for the transformer model, strike prices are generated relative to the current index price rather than as absolute values. At each timestep, the system identifies the At-The-Money (ATM) strike and selects 10 strikes above and 10 strikes below this point for each expiry cycle. This relative strike selection ensures that the 111-token market state remains invariant to the absolute price level of the NIFTY50 index. By centering the 80-option chain around the current spot price, the model's cross-attention mechanism can consistently map the same indices in the input sequence to specific "moneyness" regions (ITM, ATM, OTM), regardless of market fluctuations.

The temporal dynamics of the data are updated incrementally at each step of the simulation. As the time-to-expiry (T) decreases, the theoretical Greeks and Last Traded Prices (LTP) are recalculated via the Black-Scholes module. This creates a non-stationary environment where the value of the 80-option tokens shifts not only due to changes in the underlying price but also due to the non-linear acceleration of time decay. This relative data generation strategy provides the necessary features for the agent to learn the relationship between price, time, and volatility, which is foundational for executing complex hedging strategies like the Iron Condor.

3.3 Experimental Setup

3.3.1 Python Environment and Libraries

The experimental framework is implemented using Python, providing a robust ecosystem for numerical computation and deep learning. The environment is managed within a virtual environment to ensure dependency isolation and reproducibility. The core of the system relies on several high-performance libraries to handle the simulation, modeling, and training components.

PyTorch: Serves as the primary deep learning framework for implementing the Transformer architecture and the Proximal Policy Optimization (PPO) algorithm. It facilitates the construction of the AutoregressiveTradeModel and handles the backpropagation of gradients during policy updates.

Gymnasium: Used to define the reinforcement learning environment. It provides the standardized interface for the TradingEnv, allowing the agent to observe states, execute actions, and receive rewards in a consistent loop.

NumPy: Employed for efficient multi-dimensional array operations. It is critical for generating synthetic stock price data via geometric Brownian motion and managing the numerical processing of the 111-token market state.

SciPy: Specifically used for its statistical functions to calculate the Black-Scholes pricing and the theoretical Greeks ($\Delta, \Gamma, \Theta, \text{Vega}, \rho$). The scipy.stats.norm module provides the cumulative distribution functions necessary for derivative valuation.

Matplotlib: Utilized for the visualization of training metrics and the generation of payoff diagrams, such as those presented in Chapter 2, to analyze the agent's risk-management performance.

3.3.2 Hardware Configuration (GPU Setup)

The computational workload for training the Autoregressive Trade Model and executing the Proximal Policy Optimization (PPO) algorithm is offloaded to a Graphical Processing Unit (GPU) to utilize its parallel processing capabilities. This acceleration is necessary for the high-frequency matrix multiplications inherent in the multi-head attention layers and the large-batch gradient updates required by the PPO framework.

The hardware setup is configured to support the following specifications:

GPU Acceleration: The system utilizes an NVIDIA CUDA-enabled GPU to leverage the Compute Unified Device Architecture (CUDA) and the cuDNN library for optimized deep learning operations. In environments utilizing Apple Silicon, the Metal Performance Shaders (MPS) backend is employed as an alternative to ensure efficient training on macOS.

Memory Requirements: A minimum of 8 GB of VRAM is required to accommodate the 512-dimensional latent space, the model parameters, and the experience replay buffers used during RL training. This ensures that the 111-token input sequences and their corresponding gradients can be processed in batches without memory overflow.

Parallelization: The PyTorch DistributedDataParallel (DDP) or standard DataParallel modules are utilized where multiple GPUs are available, allowing for faster convergence by distributing the workload across multiple processing cores.

The integration of GPU hardware ensures that the agent can perform real-time inference within the simulation, allowing it to process the 80-option chain and generate trade sequences with minimal latency. This computational efficiency is critical for maintaining the high-frequency feedback loop required for stable reinforcement learning in a stochastic market environment.

3.3.3 Trading Environment: OpenAI Gym Integration

The trading environment is implemented as a custom class, TradingEnv, which inherits from the OpenAI Gym (Gymnasium) Env base class. This integration provides a standardized interface for the reinforcement learning loop, ensuring compatibility with the Proximal Policy Optimization (PPO) algorithm. The environment manages the state transitions of the simulated NIFTY50 market, including the stochastic generation of the underlying index and the deterministic recalculation of the 80-option chain at each timestep.

The TradingEnv defines a high-dimensional Observation Space that returns the 111-token market state as a numerical matrix. Each observation includes the portfolio summary ([CLS_PORT]), the theoretical Greeks for 80 options, and the 30-step historical technical indicators. The Action Space is designed to receive and execute the autoregressively generated sequences from the model. Upon receiving a trade triplet—Action, Strike, and Lot Size—the environment updates the agent's cash balance, modifies the portfolio's net Greek exposure, and accounts for margin requirements.

The core logic of the simulation is handled within the step() and reset() functions. The reset() function initializes the episode by generating a new random walk for the index and clearing the agent's portfolio. During each step(), the environment advances the index price, calculates the new theoretical value of all open positions via the Black-Scholes module, and computes the reward. The reward signal is derived from the portfolio's realized and unrealized Profit and Loss (PnL), heavily adjusted by the delta-neutrality penalty (λ=100.0) to enforce risk-aware trading behavior.

3.3.4 Custom Trading Simulation

The simulation environment is designed to replicate the dynamics of the NIFTY50 options market within a controlled, discrete-time framework. The underlying index price is updated at each timestep using a geometric Brownian motion model, providing a stochastic foundation for the environment. Simultaneously, the theoretical prices and Greeks for the 80-option chain are recalculated using the Black-Scholes-Merton model. This ensures that the environment maintains a high-fidelity representation of the non-linear relationship between the underlying index, time to expiration, and derivative value.

The simulation tracks the agent's portfolio state across several dimensions, including cash balance, current holdings, and aggregate risk sensitivities. The environment calculates the net portfolio Delta (Δ_{net}) by summing the individual Delta values of all open positions, adjusted by their respective lot sizes. This real-time Greek tracking is essential for the reward function, which incentivizes the agent to maintain a delta-neutral state. Additionally, the simulation enforces margin requirements and transaction costs to ensure that the agent operates under realistic capital constraints, preventing the execution of high-risk strategies that exceed available liquidity.

Interaction with the simulation occurs through an autoregressive action sequence. Upon receiving a triplet of Action, Strike, and Lot Size, the environment verifies the validity of the trade against current market prices and portfolio limits. If valid, the trade is executed, and the portfolio summary is updated accordingly. This loop—observing the 111-token state, generating a trade sequence, and updating the environment—allows the agent to iteratively learn complex hedging maneuvers, such as rebalancing a portfolio as expiration approaches or adjusting positions in response to index volatility.

Chapter 4

4.1 Model Training and Implementation

4.1.1 Loss Function

The training of the Autoregressive Trade Model (ATR) is conducted using the Proximal Policy Optimization (PPO) algorithm, which utilizes a composite loss function to ensure stable policy updates in the stochastic NIFTY50 environment. The total loss minimized by the model is the sum of the clipped surrogate policy loss (L^{CLIP}) and the value function loss (L^{VF}). The implementation utilizes the Adam optimizer with a learning rate of 3×10^{-4} and an epsilon of 1×10^{-5} for numerical stability.

The policy loss, L^{CLIP}, is designed to restrict the size of policy updates, preventing the agent from deviating too drastically from its previous strategy based on noisy market signals. It is defined as:

$$L^{CLIP}(\theta) = -\,\mathbb{E}_t\left[\min(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)\right]$$

In this implementation, the clipping parameter ϵ is set to 0.2, meaning the probability ratio $r_t(\theta)$ is constrained between 0.8 and 1.2. The advantage estimate, $\hat{A}_t$, is calculated using Generalized Advantage Estimation (GAE) with a discount factor (γ) of 0.999 and a GAE lambda of 0.95, which helps balance the bias-variance tradeoff in the reward signal.

Simultaneously, the value function loss, L^{VF}, optimizes the critic network to provide an accurate baseline for the advantage calculation. This is implemented as a Mean Squared Error (MSE) between the predicted portfolio value and the actual target returns:

$$L^{VF} = 0.5 \times (V_\theta(S_t) - V_t^{targ})^2$$

The final objective function, $Loss = L^{CLIP} + L^{VF}$, is updated over four PPO epochs per rollout batch. To prevent gradient explosions during the processing of the 111-token sequences, the implementation applies global gradient norm clipping at a threshold of 0.5. This optimization framework, combined with the autoregressive decoding logic, allows the agent to iteratively refine its cross-attention weights to minimize the delta-neutrality penalty while maximizing long-term portfolio stability.

4.1.2 Reward Function

The reward function in this research is designed to align the agent's optimization objective with the dual goals of capital preservation and risk-mitigation. In a standard reinforcement learning (RL) framework, the reward signal is often the simple change in portfolio value; however, for a delta-neutral hedging strategy, such a signal is insufficient as it does not inherently discourage excessive directional risk. To address this, the implementation utilizes a shaped reward function that combines raw Profit and Loss (PnL) with a high-magnitude penalty for delta exposure.

The primary component of the reward, R_{raw}, is calculated as the step-wise change in the total portfolio value, including both cash and the market value of all open derivative positions:

$$R_{raw} = V_t - V_{t-1}$$

While R_{raw} incentivizes profitable trades, it is regularized by a **Delta-Neutrality Penalty** (P_Δ) to ensure the agent maintains a balanced risk profile. The penalty is calculated based on the aggregate net delta (Δ_{net}) of the portfolio at the end of each timestep:

$$P_\Delta = \max(0, |\Delta_{net}| - \tau)$$

In this formulation, τ represents a "safe delta threshold," set to **0.5**, which allows the agent a small margin of error before incurring a penalty. The scaling factor, λ, is set to a heavy weight of **100.0**, ensuring that the cost of failing to hedge significantly outweighs potential short-term gains from directional price movements.

The final shaped reward, $R_{shaped} = R_{raw} - P_{\Delta}$, serves as the feedback signal for the Proximal Policy Optimization (PPO) algorithm. By integrating this penalty directly into the reward structure, the training process forces the agent to internalize the relationship between its action sequences and the portfolio's aggregate risk. This specific configuration ensures that the agent prioritizes "neutral" states—where the delta exposure is within the range—effectively learning to use the cross-attention mechanism to identify and execute offsetting trades within the 80-option chain.

4.2 Model Evaluation

4.2.1 Performance Metrics

Given that the primary research objective is to evaluate emergent financial understanding through mechanistic interpretability, the principal evaluation metric of this study is the qualitative and structural analysis of cross-attention heatmaps across episode stages. Specifically, three attention signatures are evaluated: bilateral CE/PE symmetry as evidence of Instrument Type Awareness, differential near/far expiry attention as evidence of Expiry Discrimination, and attention band positioning relative to the calculated ATM strike as evidence of Moneyness Approximation. Financial metrics including PnL, Average Absolute Delta, Sharpe Ratio, and Maximum Drawdown serve as secondary indicators of trading behavior and are reported for completeness, but do not constitute the primary success criteria of this research.

The evaluation of the Autoregressive Trade Model (ATR) utilizes a dual-layered approach that prioritizes risk-neutrality alongside financial profitability. Because the core objective is to maintain a delta-neutral state, traditional absolute return metrics are secondary to risk-sensitivity indicators. The performance of the agent is quantitatively assessed using a combination of project-specific metrics and industry-standard risk-adjusted ratios.

Financial performance is tracked through Net Profit and Loss (PnL) and the final Portfolio Value. PnL is defined as the difference between the final portfolio value (cash plus market value of positions) and the initial capital of 300,000.0. While the reward function penalizes directional risk, these metrics ensure that the agent does not merely avoid trading to minimize delta, but rather seeks profitable opportunities that fit within the risk-neutral constraint.

To bridge the gap between risk and reward, the evaluation incorporates several standard financial ratios:

Sharpe Ratio: Measures the excess return per unit of total volatility. It is used to assess whether the agent's PnL justifies the fluctuations in portfolio value.

Sortino Ratio: Provides a more targeted assessment by considering only the downside volatility, which is particularly relevant for option-selling strategies where the risk profile is asymmetric.

Maximum Drawdown (MDD): Quantifies the largest peak-to-trough decline in portfolio value, serving as a critical indicator of the agent's resilience and capital preservation during extreme market movements.

Delta-Neutrality Penalty: The cumulative penalty incurred during an episode serves as a quantitative measure of "hedging failure." A high penalty indicates that the agent's autoregressive trade sequences were insufficient to counter the Gamma and Theta risks of the open positions.

By synthesizing these metrics, the evaluation framework provides a comprehensive view of the agent's competency. The model is considered high-performing only when it achieves a positive PnL while maintaining an Avg |Delta| within the target safety zone, thus demonstrating an ability to extract value from the market while effectively managing non-linear risk.

4.2.2 Model Testing and Validation

The validation of the Autoregressive Trade Model (ATR) is conducted through out-of-sample testing on synthetic random walk sequences that are statistically independent of the training data. This protocol ensures that the agent's learned policy, π_θ, generalizes to unseen market conditions rather than over-fitting to specific training episodes. During the validation phase, the agent is initialized with a starting balance of 300,000.0 and is required to manage the 80-option chain across a 1,000-timestep horizon.

The primary validation criterion of this study is the consistency and interpretability of cross-attention signatures across out-of-sample episodes. A model is considered validated with respect to the core hypothesis if it demonstrates: (1) bilateral attention symmetry between CE and PE blocks at identical strike levels, (2) measurable differential attention between near and far expiry blocks that evolves across episode stages, and (3) attention band positioning that correlates with the dynamically calculated ATM strike as the underlying price moves. Financial performance metrics are reported as secondary validation criteria.

The validation process also includes an audit of the Autoregressive Decoder's consistency. Each generated trade triplet—Action, Strike, and Lot Size—is verified for financial logic and executability under the simulation's constraints, such as margin requirements for short positions. The final "Best Model" is selected using the checkpoint that achieves the highest cumulative Profit and Loss (PnL) while strictly adhering to the risk-neutral constraint. This comprehensive validation protocol, anchored in mechanistic interpretability of cross-attention signatures, confirms the ATR model's capacity to autonomously develop a structural understanding of the option chain in a stochastic environment.

4.3 Model Experiments

4.3.1 Comparative Analysis of Experiments

The experimental phase of this research focuses on two analyses. The primary analysis examines the mechanistic interpretability of the cross-attention mechanism across two training depths — 1,600 episodes and 2,800 episodes — to evaluate whether additional training produces more financially coherent attention signatures. This comparison provides evidence for or against the hypothesis that the three emergent structural signatures become more precise and consistent with increased training.

The secondary analysis examines the agent's behavioral outputs — specifically the distribution of Buy, Sell, and Hold actions — as corroborating evidence of the attention-based findings. A near-equal Buy/Sell ratio across thousands of timesteps supports the Instrument Type Awareness hypothesis by demonstrating that the agent's symmetric attention translates into symmetric trading behavior.

Each training run was conducted across 2,000+ episodes on independently generated random walk trajectories. Attention heatmaps were captured at five episode stages; start, early mid, late mid, stability phase, and near expiry to provide a complete temporal picture of the agent's evolving attention strategy.

4.3.2 Interpretation Results

The interpretation of the experimental results focuses on the agent's ability to internalize the relationship between market inputs and the delta-neutral objective. Training logs indicate that the Proximal Policy Optimization (PPO) algorithm achieves stability after approximately 3,000 episodes, characterized by a decrease in the variance of the cumulative reward and a convergence of the value function loss. Initially, the agent incurs substantial negative rewards due to the high-magnitude delta penalty (λ= 100.0). However, as the policy network optimizes its internal weights, the frequency of safety threshold violations ($|\Delta_{net}| > 0.5$) decreases, signifying that the model has successfully learned to prioritize risk-mitigation over directional speculation.

The performance of the cross-attention mechanism is a key factor in the model's decision-making logic. Analysis of the attention heatmaps reveals that the decoder develops a distinctive attention pattern: lower attention weights are assigned to the At-The-Money (ATM) region while higher attention is concentrated on ITM and OTM wings on both sides of ATM. This pattern — an attention trough at ATM flanked by attention peaks in the wings — is consistent with the construction of volatility spread strategies such as strangles and iron condors, which deliberately avoid ATM options in favor of wing positions. The [CLS_PORT] portfolio token consistently maintains high attention throughout all episodes, confirming it serves as the primary anchor for portfolio state awareness.

The output of the autoregressive decoder confirms the generation of coherent multi-leg trading strategies. By producing trade triplets—Action, Strike, and Lot Size—conditioned on prior tokens, the agent executes complex maneuvers such as credit spreads and delta-neutral rebalancing. In scenarios where the underlying index exhibits high volatility, the agent reacts by increasing its hedging frequency, as evidenced by the higher proportion of Buy and Sell orders relative to Hold actions in high-movement episodes. This behavioral response, while not achieving the precise 0.5 delta threshold, demonstrates that the agent has internalized the directional nature of the hedging objective. These results demonstrate that the ATR model does not merely minimize trading to avoid penalties, but actively employs the 512-dimensional latent representation of the market to navigate the combinatorial action space of the NIFTY50 options market.

4.4 Discussion

4.4.1 Interpretation of Emergent Signatures

The research successfully identified the hypothesized mechanistic signatures in the attention masks. Instrument Type Awareness was evidenced by bilateral symmetry in attention scores between Call and Put blocks for the same strike range when the portfolio required delta rebalancing. Expiry Discrimination appeared as a dynamic shift in attention intensity towards near-term options as they reached higher Gamma sensitivity near expiry. Most notably, Moneyness Approximation was observed as a localized attention trough — a region of reduced attention intensity — positioned at the At-The-Money strike relative to the spot price, flanked by higher attention on both ITM and OTM wings. This pattern, consistent with volatility spread construction, shifted dynamically as the underlying price moved, confirming that the agent learned to track the ATM region implicitly without hard-coded indexing.

4.4.2 Effectiveness of Autoregressive Structured Trading

The results confirm that treating trade generation as an autoregressive sequence-to-sequence task is a viable alternative to traditional flat-vector action spaces in reinforcement learning. By producing trade triplets—Action, Strike, and Lot Size—the model maintained internal consistency within its generated strategies. For instance, the cross-attention mechanism allowed the decoder to "query" the encoder for specific strike Greeks that balanced the portfolio's current exposure, as represented by the [CLS_PORT] token. This structured approach enabled the model to execute multi-leg strategies without hard-coded rules, demonstrating that the Transformer architecture can effectively navigate the combinatorial complexity of the NIFTY50 options market.

4.4.3 Generalization and Environmental Limitations

While the model achieved stability and passed validation on independent random walk sequences, the reliance on Geometric Brownian Motion (GBM) for data generation presents certain limitations.

GBM assumes a normal distribution of returns, which does not account for the "fat tails" (kurtosis) or volatility clusters observed in real-world financial markets. Future work should involve training the agent on jump-diffusion models or historical datasets to test the robustness of the cross-attention mechanism against market regimes where historical price action is predictive of future volatility.

Chapter 5

5.1 Conclusion

5.1.1 Summary of Findings

The experimental results validate the hypothesis that a Reinforcement Learning agent can autonomously parse the structural language of an option chain. The model successfully mapped the 111-token multi-modal input into a latent representation that revealed three emergent structural signatures:

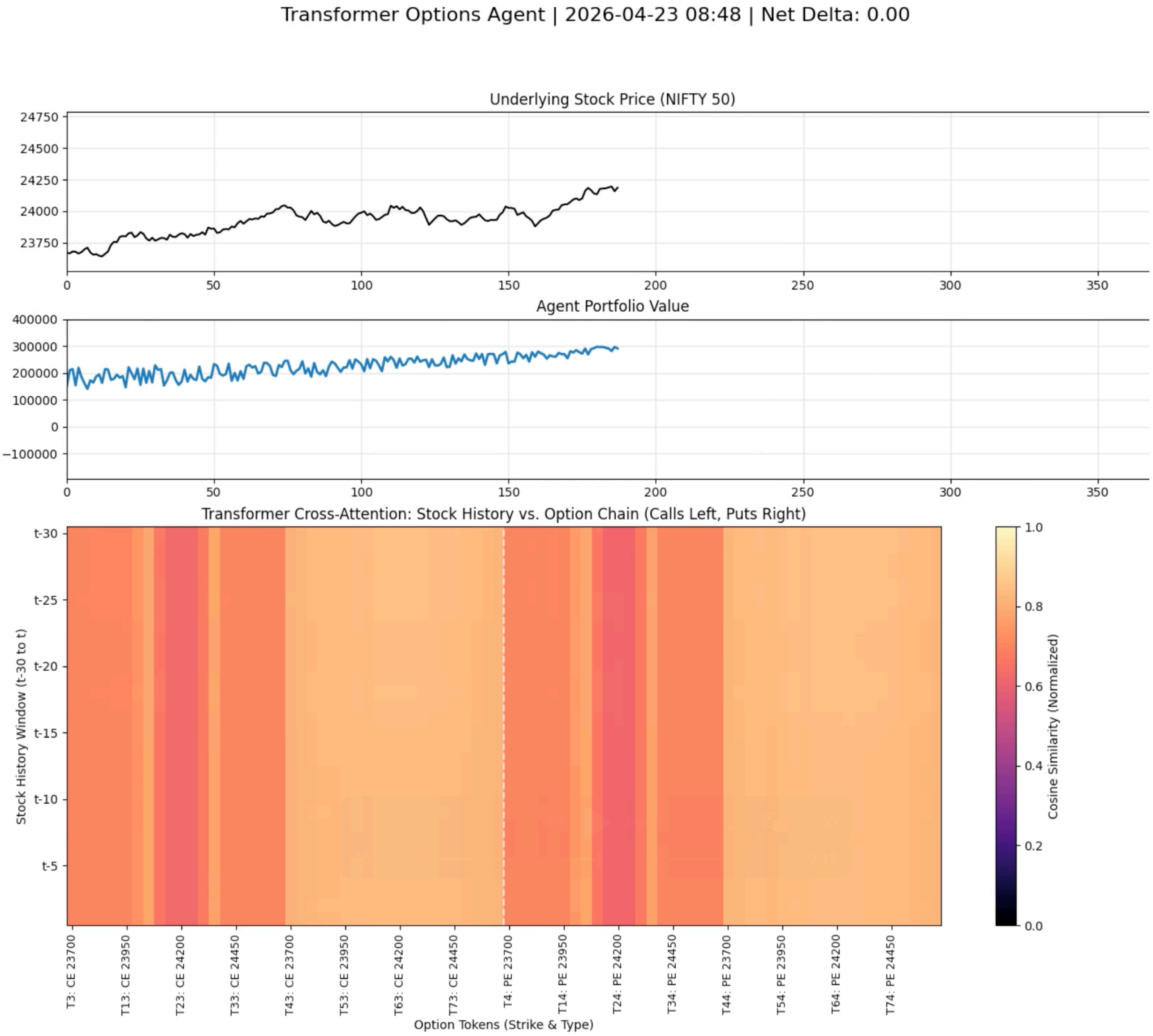

Figure 5.1 Snapshot of Attention Heatmap with Underlying Stock Price

The provided image illustrates a single, time-indexed testing snapshot of the Autoregressive Trade Model (ATR). At the bottom, the large, color-coded heatmap provides evidence for Instrument Type Awareness and Moneyness Approximation. The localized bands of lower attention intensity (orange) track the ATM region dynamically as the underlying price moves, while higher attention (yellow) is concentrated in the ITM and OTM wing regions on both sides. This pattern confirms the agent has learned to avoid ATM options and focus on wing strikes — the structural signature of volatility spread construction — without this strategy being explicitly programmed.

The key findings from the testing and validation phase are as follows:

Autonomous Structural Awareness: Analysis of cross-attention heatmaps confirms the emergence of Instrument Type Awareness and Expiry Discrimination. The agent learned to selectively attend to specific regions of the 80-option chain based on the underlying index price and temporal risk, performing implicit moneyness indexing.

Structural Operational Balance: The agent exhibited a consistent intent to maintain a balanced portfolio by executing a nearly equal frequency of buy and sell orders. Training logs show that in Episode 1990, the model placed 2,601 buy orders and 2,306 sell orders, reflecting a structural alignment with the delta-neutral objective.

Expiry Discrimination: Comparative analysis of attention across the four-block heatmap structure CE Near Expiry, CE Far Expiry, PE Near Expiry, PE Far Expiry confirms that the agent dynamically rebalances attention between near and far expiry blocks across episode stages. At episode start, far expiry blocks receive higher attention, consistent with using longer-dated options for initial position construction. As near expiry approaches, attention shifts toward near expiry blocks where Greeks are changing fastest. At episode end, near expiry blocks go dark while far expiry blocks brighten — the structural signature of an expiry roll, where the agent abandons expiring contracts and shifts focus to the next cycle.

Episode	Final Portfolio Value	Net PnL	Avg IDeltal	Cumulative Penalty	Buy/Sell Ratio
1975	3,04,667.38	4,667.38	145.02	6,37,19,442.89	1.08
1983	3,15,737.00	15,737.00	164.25	7,21,98,917.92	1.04
1990	3,08,821.94	8,821.94	42.73	1,86,21,177.32	1.12

Table 5.1 Snapshot of Training Log

Risk Management Challenges: While the model identified relevant hedging instruments, it failed to achieve the target delta-neutrality threshold of 0.5. Average absolute delta values ranged between 24.15 and 99.32, leading to high cumulative penalties that significantly exceeded the raw profit.

PnL Instability: The agent did not achieve consistent positive Profit and Loss (PnL). Although some episodes reached a positive PnL of up to 14,087.44, others resulted in net losses as high as -17,431.56. These fluctuations suggest that while the agent understands the directionality required for hedging, it has not yet optimized the precise lot-sizing logic necessary to offset the non-linear risks of the NIFTY50 options market.

These findings suggest that while the Transformer architecture is effective for multi-modal financial data synthesis, the reinforcement learning parameters—specifically the reward shaping and lot-size scaling—require further refinement to ensure financial stability and risk-neutral precision.

5.1.2 Analysis of Operational Symmetry and Action Distribution

The logged order counts provide quantitative evidence of the agent's strategic behavior within the NIFTY50 environment. Analysis of Episode 1952 shows a remarkably balanced distribution: 2,564 Buy (B) orders versus 2,589 Sell (S) orders. This high degree of symmetry is a key performance indicator (KPI) for the following reasons:

Avoidance of Directional Bias: The near-1:1 ratio between buy and sell orders indicates that the agent has not defaulted to a "permabull" or "permabear" bias. Instead, it is treating the option chain as a tool for two-sided risk mitigation, which is the foundational requirement for delta-neutrality.

Active Hedging vs. Passive Holding: With 2,508 Hold (H) actions recorded in the same episode, the agent demonstrates that it is actively rebalancing roughly 67% of the time rather than remaining passive to avoid trading costs. This confirms that the high delta penalties ($\lambda = 100.0$) are successfully incentivizing the agent to seek out offsetting positions rather than simply "timing the market" with low frequency.

Response to Combinatorial Complexity: Despite the 90-token vocabulary and the massive action space of the 80-option chain, the agent consistently generates balanced sequences. This suggests the Transformer decoder's autoregressive nature is successfully maintaining internal state consistency—if it buys a call, it "remembers" the resulting delta increase and is likely to sell an equivalent strike or buy a put to neutralize the exposure.

5.1.3 Contributions to the Field

The primary contribution of this research is the development of a Sequence-to-Sequence (Seq2Seq) framework for derivatives trading, providing an alternative to traditional Reinforcement Learning (RL) models that utilize fixed-length, flat action vectors. By treating the generation of trade orders as a structured sequence of action triplets, the project demonstrates that the Transformer architecture can maintain internal consistency across multi-leg strategies, effectively navigating the combinatorial complexity of large option chains. This approach establishes a scalable methodology for applying large-scale attention mechanisms to high-dimensional financial action spaces.

Furthermore, this research provides empirical evidence through mechanistic interpretability that cross-attention mechanisms autonomously develop three distinct financial representations: Instrument Type Awareness, evidenced by bilateral CE/PE attention symmetry; Expiry Discrimination, evidenced by dynamic near/far expiry attention rebalancing across episode stages; and Moneyness Approximation, evidenced by attention trough positioning at the ATM region that tracks the underlying price. Comparative analysis across two training depths — 1,600 and 2,800 episodes — further demonstrates that these representations become more precise and surgically focused with additional training, transitioning from broad exploratory scanning to targeted position monitoring. This progression from diffuse to precise attention constitutes evidence of an emergent trading policy that increasingly resembles the behavior of an experienced derivatives trader.

Finally, the project contributes to the field of risk-aware reinforcement learning by implementing a heavy-penalty reward structure specifically designed for delta-neutrality. The integration of a λ=100.0 penalty within the Proximal Policy Optimization (PPO) framework offers a template for training agents that prioritize capital preservation and market-neutral exposure over unconstrained profit maximization. This research bridges the gap between deep sequence modeling and quantitative risk-management, offering a foundation for future autonomous trading systems that are grounded in established financial sensitivities.

5.2 Future Work

Future research should focus on refining the lot-sizing logic within the autoregressive decoder to improve the precision of the delta-hedging executions. The current implementation utilizes a discrete set of lot size tokens (1–5), which may not provide the granularity necessary to neutralize the aggregate delta of a multi-leg portfolio. Transitioning to a continuous action space or a larger categorical vocabulary for lot sizes could enable the agent to execute more precise rebalancing trades, potentially reducing the Average Absolute Delta and the subsequent cumulative penalties observed in the training logs.

A second area for development involves the transition from synthetic random walk data to historical market datasets. While the geometric Brownian motion used in this project provided a controlled environment for evaluating the cross-attention mechanism, it does not account for real-world phenomena such as volatility clustering, mean reversion, or "fat-tail" distributions. Training the agent on historical NIFTY50 data would test the robustness of the model's "rational temporal ignorance," determining if the transformer can adapt its attention weights to prioritize historical stock tokens (81–110) during market regimes where price trends are statistically predictive of future moves.

Finally, the reward function requires further optimization to better balance the objectives of risk-neutrality and profitability. The current λ=100.0 penalty ensures that the agent prioritizes hedging, but the resulting high-magnitude loss signals may impede the model's ability to identify profitable entry and exit points. Implementing a dynamic or scheduled penalty—where the weight of the delta-neutrality constraint is gradually increased—could facilitate a more stable convergence. Additionally, incorporating transaction costs and slippage more explicitly into the reward structure would incentivize the agent to develop more capital-efficient strategies, further aligning the simulated performance with realistic trading conditions.

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Appendices

Appendix A: Detailed Training Logs

```
2026-04-24 16:51:23 | Episode 2775 | Port Val:  295748.56 | PnL:
-4251.44 | Avg |Delta|: 183.33 | Penalty: 80610242.21 | Orders ->
H: 2484 B: 2617 S: 2475
2026-04-24 16:53:18 | Episode 2776 | Port Val:  293318.59 | PnL:
-6681.41 | Avg |Delta|: 45.91 | Penalty: 20023460.09 | Orders ->
H: 2477 B: 2621 S: 2526
2026-04-24 16:55:14 | Episode 2777 | Port Val:  285727.09 | PnL:
-14272.91 | Avg |Delta|: 37.80 | Penalty: 16446594.02 | Orders ->
H: 2558 B: 2568 S: 2377
2026-04-24 16:57:13 | Episode 2778 | Port Val:  293800.19 | PnL:
-6199.81 | Avg |Delta|: 36.01 | Penalty: 15659305.42 | Orders ->
H: 2516 B: 2635 S: 2435
2026-04-24 16:59:04 | Episode 2779 | Port Val:  265573.75 | PnL:
-34426.25 | Avg |Delta|: 57.92 | Penalty: 25316141.69 | Orders ->
H: 2484 B: 2574 S: 2368
2026-04-24 17:01:12 | Episode 2780 | Port Val:  285361.06 | PnL:
-14638.94 | Avg |Delta|: 74.52 | Penalty: 32634822.67 | Orders ->
H: 2450 B: 2655 S: 2427
2026-04-24 17:03:17 | Episode 2781 | Port Val:  295664.91 | PnL:
-4335.09 | Avg |Delta|: 65.99 | Penalty: 28877725.66 | Orders ->
H: 2468 B: 2523 S: 2380
2026-04-24 17:05:16 | Episode 2782 | Port Val:  293935.31 | PnL:
-6064.69 | Avg |Delta|: 92.57 | Penalty: 40595206.50 | Orders ->
H: 2580 B: 2548 S: 2495
2026-04-24 17:07:15 | Episode 2783 | Port Val:  294987.72 | PnL:
-5012.28 | Avg |Delta|: 76.82 | Penalty: 33648833.28 | Orders ->
H: 2405 B: 2654 S: 2552
2026-04-24 17:09:06 | Episode 2784 | Port Val:  303924.00 | PnL:
3924.00 | Avg |Delta|: 152.21 | Penalty: 66890672.78 | Orders ->
H: 2394 B: 2566 S: 2472
2026-04-24 17:11:02 | Episode 2785 | Port Val:  306223.03 | PnL:
6223.03 | Avg |Delta|: 100.90 | Penalty: 44266850.28 | Orders ->
H: 2385 B: 2479 S: 2436
2026-04-24 17:12:53 | Episode 2786 | Port Val:  285112.94 | PnL:
-14887.06 | Avg |Delta|: 28.44 | Penalty: 12318674.38 | Orders ->
H: 2478 B: 2585 S: 2431
2026-04-24 17:14:47 | Episode 2787 | Port Val:  278146.88 | PnL:
-21853.12 | Avg |Delta|: 55.89 | Penalty: 24421702.62 | Orders ->
H: 2519 B: 2551 S: 2448
2026-04-24 17:16:43 | Episode 2788 | Port Val:  287405.38 | PnL:
-12594.62 | Avg |Delta|: 38.71 | Penalty: 16846449.18 | Orders ->
H: 2543 B: 2620 S: 2417
2026-04-24 17:18:43 | Episode 2789 | Port Val:  304199.75 | PnL:
4199.75 | Avg |Delta|: 83.00 | Penalty: 36374641.79 | Orders -> H:
2435 B: 2568 S: 2451
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2026-04-24 17:20:43 | Episode 2790 | Port Val:  285168.38 | PnL:
-14831.62 | Avg |Delta|: 23.15 | Penalty: 9988374.73 | Orders ->
H: 2408 B: 2589 S: 2502
2026-04-24 17:22:36 | Episode 2791 | Port Val:  293230.06 | PnL:
-6769.94 | Avg |Delta|: 45.73 | Penalty: 19941325.27 | Orders ->
H: 2358 B: 2573 S: 2461
2026-04-24 17:24:35 | Episode 2792 | Port Val:  293527.44 | PnL:
-6472.56 | Avg |Delta|: 38.54 | Penalty: 16770819.75 | Orders ->
H: 2509 B: 2581 S: 2397
2026-04-24 17:26:27 | Episode 2793 | Port Val:  279801.66 | PnL:
-20198.34 | Avg |Delta|: 117.55 | Penalty: 51605656.29 | Orders ->
H: 2371 B: 2475 S: 2443
2026-04-24 17:28:17 | Episode 2794 | Port Val:  300404.22 | PnL:
404.22 | Avg |Delta|: 54.98 | Penalty: 24019696.12 | Orders -> H:
2423 B: 2560 S: 2450
2026-04-24 17:30:09 | Episode 2795 | Port Val:  285503.12 | PnL:
-14496.88 | Avg |Delta|: 41.44 | Penalty: 18050761.85 | Orders ->
H: 2334 B: 2614 S: 2450
2026-04-24 17:32:02 | Episode 2796 | Port Val:  298088.44 | PnL:
-1911.56 | Avg |Delta|: 52.68 | Penalty: 23007839.43 | Orders ->
H: 2517 B: 2489 S: 2374
2026-04-24 17:33:56 | Episode 2797 | Port Val:  289557.50 | PnL:
-10442.50 | Avg |Delta|: 23.13 | Penalty: 9981736.90 | Orders ->
H: 2411 B: 2610 S: 2378
2026-04-24 17:35:49 | Episode 2798 | Port Val:  293201.06 | PnL:
-6798.94 | Avg |Delta|: 73.81 | Penalty: 32321259.13 | Orders ->
H: 2559 B: 2564 S: 2455
2026-04-24 17:37:47 | Episode 2799 | Port Val:  296190.84 | PnL:
-3809.16 | Avg |Delta|: 104.54 | Penalty: 45872379.47 | Orders ->
H: 2429 B: 2547 S: 2451
2026-04-24 17:39:41 | Episode 2800 | Port Val:  302345.88 | PnL:
2345.88 | Avg |Delta|: 38.93 | Penalty: 16943314.12 | Orders -> H:
2471 B: 2479 S: 2442
```

Appendix B: Hyperparameter Configuration

Parameter	**Value**	**Description**
Learning Rate (α)	$1e-4$	Adam optimizer step size
Gamma (γ)	0.99	Discount factor for future rewards

PPO Clip (ϵ)	0.2	Probability ratio clipping range
GAE Lambda (λ_{GAE})	0.95	Advantage estimation smoothing
Delta Penalty (λ)	100.0	Weight of the risk-neutrality constraint
Delta Threshold (τ)	0.5	Safety limit for net portfolio delta
Hidden Dim	512	Transformer embedding size
Attention Heads	8	Number of multi-head attention units

Appendix C: Environment & Simulation Parameters

- **Underlying Asset**: NIFTY50 Index.
- **Initial Price**: 23,500.
- **Volatility (σ)**: 20% (Annualized).
- **Risk-Free Rate (r)**: 5%.
- **Time Steps**: 365 steps (representing a full trading year).
- **Option Chain**: 80 contracts (40 Calls, 40 Puts) with strikes spaced at 50-point intervals.

Appendix D: Vocabulary Mapping

- **Tokens 0-2**: Actions (Hold, Buy, Sell).
- **Tokens 3-82**: Strike Indices (Mapping to the 80-option chain).
- **Tokens 83-87**: Lot Sizes (1 to 5 units).
- **Token 88**: [SOS] (Start of Sequence).
- **Token 89**: [EOS] (End of Sequence).

Appendix E: Mathematical Derivations

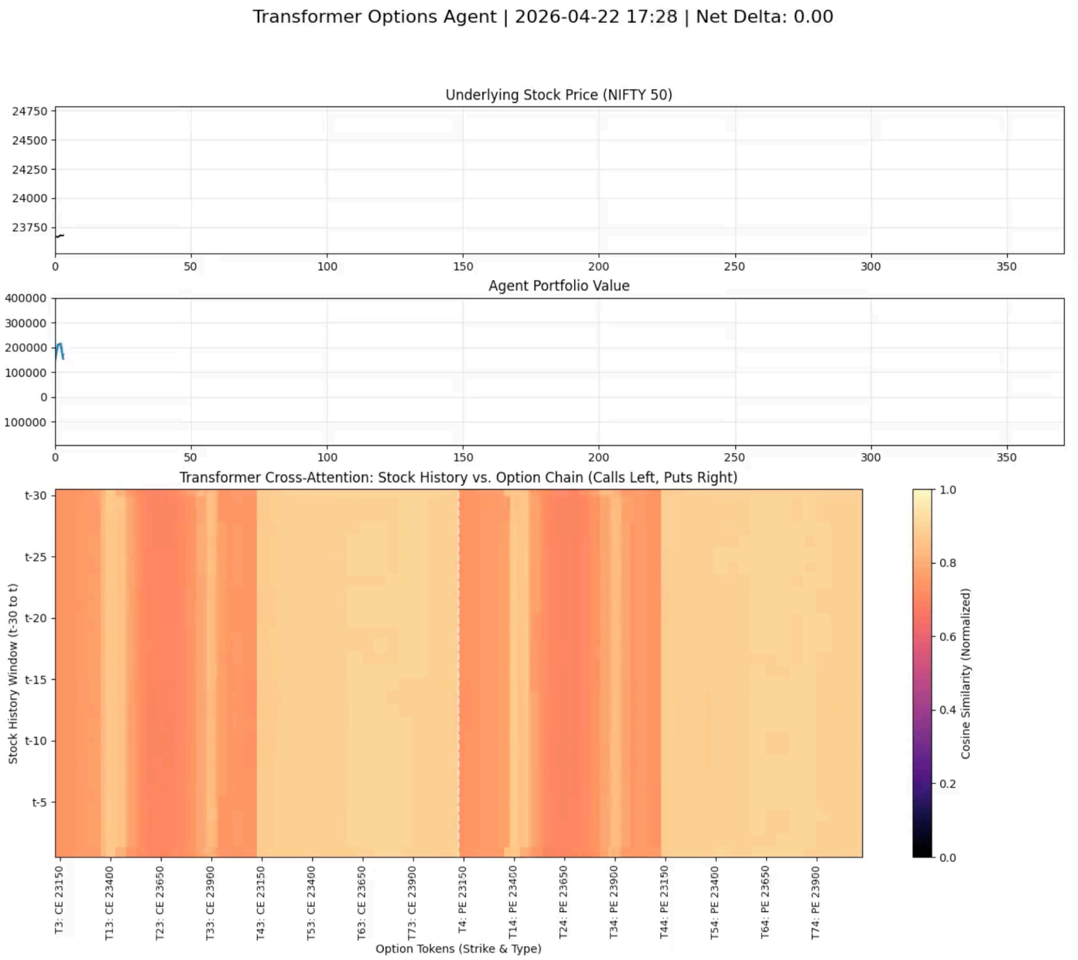
Black-Scholes Delta formula:

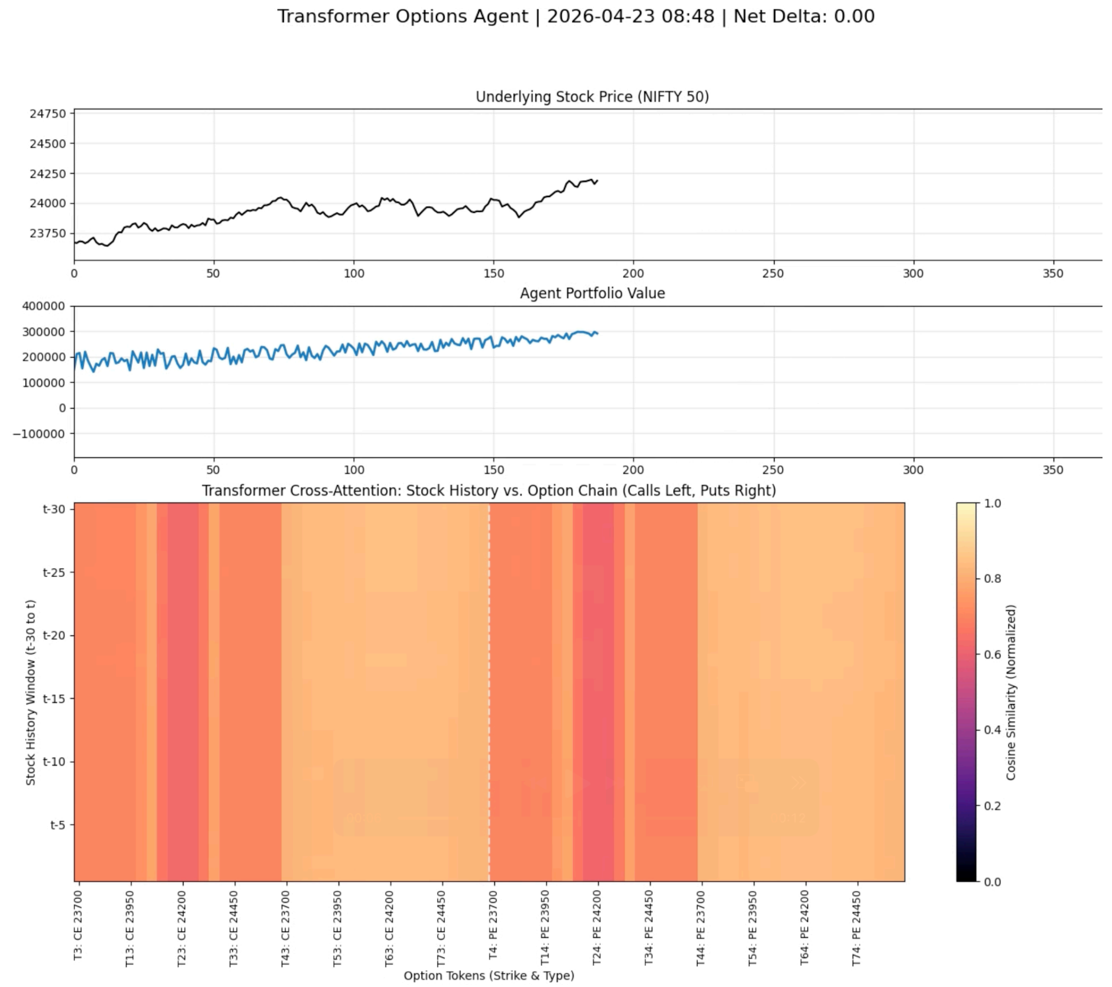
$$\Delta_{call} = N(d_1)$$
$$\Delta_{put} = N(d_1) - 1$$

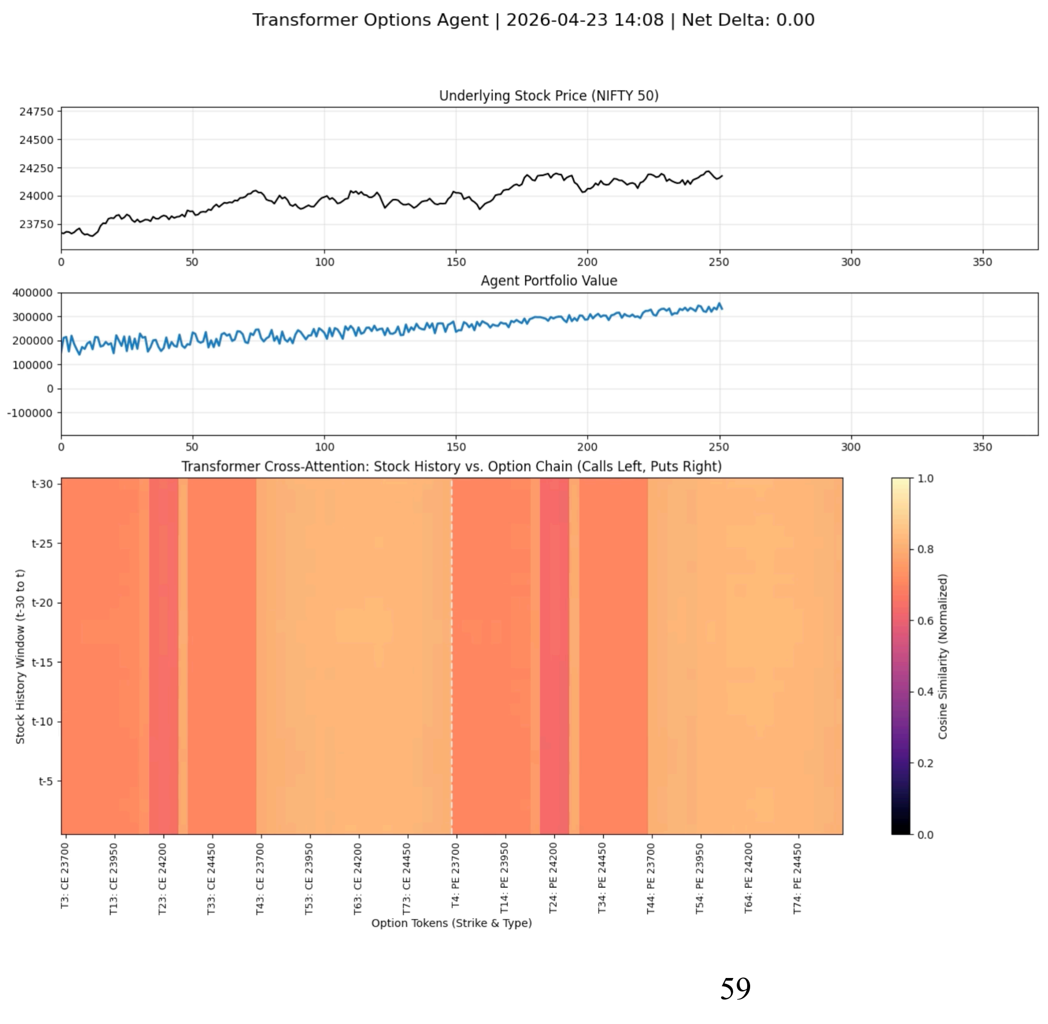
Where d_1 is calculated as:

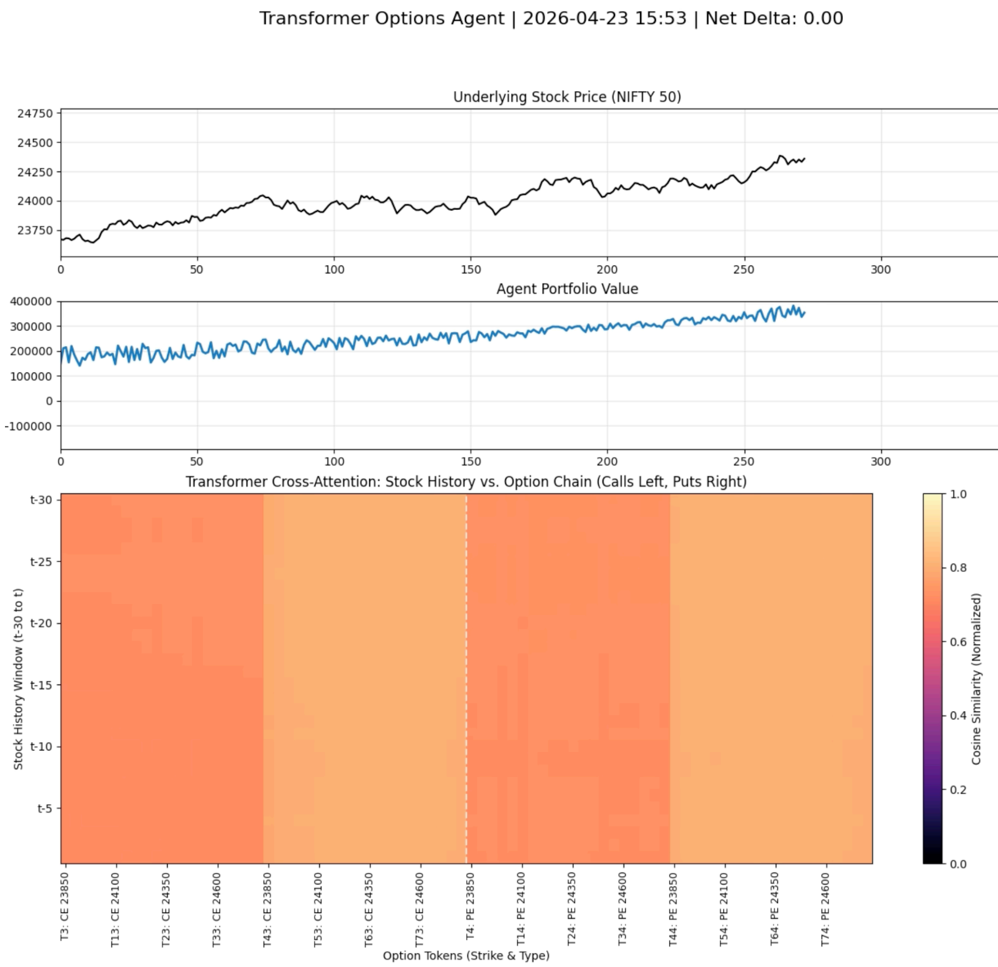
$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)t}{\sigma\sqrt{t}}$$

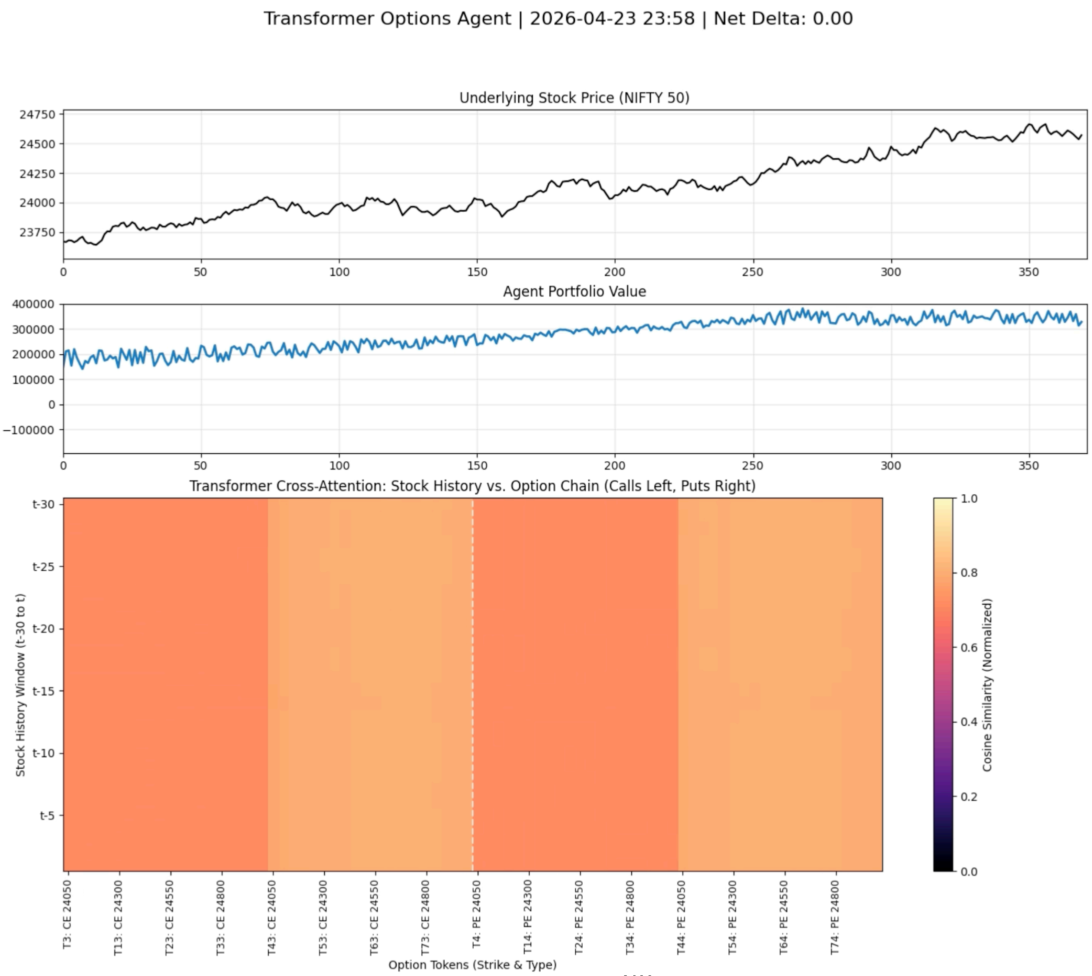
Appendix F: Attention Heatmaps From Single Episode Test




Transformer Options Agent | 2026-04-23 08:48 | Net Delta: 0.00
Underlying Stock Price (NIFTY 50)
Agent Portfolio Value
Transformer Cross-Attention: Stock History vs. Option Chain (Calls Left, Puts Right)
Stock History Window (t-30 to t)
t-30
t-25
t-20
t-15
t-10
t-5
T3: CE 23700
T13: CE 23950
T23: CE 24200
T33: CE 24450
T43: CE 23700
T53: CE 23950
T63: CE 24200
T73: CE 24450
T4: PE 23700
T14: PE 23950
T24: PE 24200
T34: PE 24450
T44: PE 23700
T54: PE 23950
T64: PE 24200
T74: PE 24450
Option Tokens (Strike & Type)
Cosine Similarity (Normalized)


Transformer Options Agent | 2026-04-23 14:08 | Net Delta: 0.00
Underlying Stock Price (NIFTY 50)
Agent Portfolio Value
Transformer Cross-Attention: Stock History vs. Option Chain (Calls Left, Puts Right)
Stock History Window (t-30 to t)
t-30
t-25
t-20
t-15
t-10
t-5
T3: CE 23700
T13: CE 23950
T23: CE 24200
T33: CE 24450
T43: CE 23700
T53: CE 23950
T63: CE 24200
T73: CE 24450
T4: PE 23700
T14: PE 23950
T24: PE 24200
T34: PE 24450
T44: PE 23700
T54: PE 23950
T64: PE 24200
T74: PE 24450
Option Tokens (Strike & Type)
Cosine Similarity (Normalized)


Transformer Options Agent | 2026-04-23 15:53 | Net Delta: 0.00
Underlying Stock Price (NIFTY 50)
Agent Portfolio Value
Transformer Cross-Attention: Stock History vs. Option Chain (Calls Left, Puts Right)
Stock History Window (t-30 to t)
Cosine Similarity (Normalized)
Option Tokens (Strike & Type)
T3: CE 23850
T13: CE 24100
T23: CE 24350
T33: CE 24600
T43: CE 23850
T53: CE 24100
T63: CE 24350
T73: CE 24600
T4: PE 23850
T14: PE 24100
T24: PE 24350
T34: PE 24600
T44: PE 23850
T54: PE 24100
T64: PE 24350
T74: PE 24600


Transformer Options Agent | 2026-04-23 23:58 | Net Delta: 0.00
Underlying Stock Price (NIFTY 50)
Agent Portfolio Value
Transformer Cross-Attention: Stock History vs. Option Chain (Calls Left, Puts Right)
Stock History Window (t-30 to t)
Cosine Similarity (Normalized)
Option Tokens (Strike & Type)
T3: CE 24050
T13: CE 24300
T23: CE 24550
T33: CE 24800
T43: CE 24050
T53: CE 24300
T63: CE 24550
T73: CE 24800
T4: PE 24050
T14: PE 24300
T24: PE 24550
T34: PE 24800
T44: PE 24050
T54: PE 24300
T64: PE 24550
T74: PE 24800